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Software Requirements Specification (SRS)

**EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM
WITH OCR LEDGER DIGITIZATION**

Version 1.0

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Revision Page

a. **Overview**

This SRS document includes an introduction to ePustaka Munshi: Smart Library with OCR Ledger Digitization and the descriptions of all the functionalities of the proposed system, that cover both functional and non-functional requirements of the system.

b. **Target Audience**

This document is targeted to business owner, stakeholder, and the developer of the system

c. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Wan Zafirzan Bin Wan Tarmizan	Initial Draft of SRS	07/01/2026

Note:

This Software Requirements Specification (SRS) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998) that is simplified and customized to meet the need of SECJ3203 FYP1 SE course at Faculty of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly. **This template is tailored for SRS in an agile-driven software development approach.**

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Appendix A: <Evidence of Requirements Elicitation Artefacts>

Appendix B: <Evidence of Requirements Elicitation Artefacts>

Update this table of content accordingly

1. Introduction

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the development of a Smart Library Management System for SMK Abdullah Munshi. The system is intended to support the digital transformation of existing manual library operations by introducing OCR-based digitization, barcode-assisted transactions, and centralized record management.

The SRS serves as a formal reference for understanding the system's purpose, scope, constraints, and expected behavior. It provides a clear specification to guide system design, development, testing, and evaluation throughout the project lifecycle.

1.1 Purpose

The purpose of this SRS document is to clearly define the requirements of the Smart Library Management System to be developed for SMK Abdullah Munshi. It outlines the system's functionalities, constraints, and operating environment to ensure that the final product meets the needs of its users and stakeholders.

This document is intended for:

- Project supervisors and examiners, to evaluate the completeness and feasibility of the system requirements
- Developers, to serve as a reference during system implementation
- Stakeholders such as librarians and student assistants, to understand the system's capabilities and limitations

This SRS describes the expected behavior of the system without detailing implementation-specific designs.

1.2 Scope

The software product is a web-based Smart Library Management System designed to support library operations at SMK Abdullah Munshi.

The system will:

- Digitize handwritten book records from existing physical ledgers using Optical Character Recognition (OCR)
- Store book, student, and transaction data in a centralized database
- Manage borrowing and returning activities using barcode scanning

- Provide role-based access for librarians and student assistants
- Generate basic reports related to book inventory and borrowing records

The system will not:

- Support mobile applications or smartphone-based scanning
- Handle online payments, fines, or external library integrations
- Integrate with national or third-party digital library systems

The application aims to improve data accuracy, reduce manual workload, preserve historical records, and streamline daily library operations. The scope is consistent with the project objectives and aligns with the operational constraints of a secondary school environment.

1.3 Definitions, Acronyms and Abbreviation

Term / Acronym	Description
SRS	Software Requirements Specification
OCR	Optical Character Recognition, a technology used to convert handwritten or printed text into digital text
SLMS	Smart Library Management System
UML	Unified Modeling Language
CRUD	Create, Read, Update, Delete
Barcode Scanner	A USB-based input device used to scan book barcodes
Ledger	A physical book used to record library information manually
Role-Based Access	Access control based on user roles such as librarian or student assistant

1.4 References

The following documents and resources were referenced during the preparation of this SRS:

- Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson Education Limited.
- IEEE Computer Society. (2014). *IEEE Standard for Software Requirements Specifications (IEEE Std 830-1998)*.
- Alkhidmah Journal. (2023). *Digital Library Management System Study*. Retrieved from <https://ejurnalqarnain.stisnq.ac.id/index.php/ALKHIDMAH/article/view/501>
- Libsys Malaysia. (n.d.). *Library Management System*. Retrieved from <https://www.libsys.my/>
- CDS Online. (n.d.). *Open Source Library Systems & Digitisation*. Retrieved from <https://cdsol.com.my/web/open-source-library-systems-digitisation/>

1.5 Overview

This SRS document is organized to provide a structured description of the Smart Library Management System. Following this introduction, subsequent sections detail the overall system description, functional and non-functional requirements, external interface requirements, and system constraints.

The document is structured to allow readers to first understand the system context and objectives before reviewing detailed requirement specifications. This organization ensures clarity and traceability between the project objectives, system scope, and implemented features.

2. Specific Requirements

2.1 User Roles

This system is designed to support multiple user roles based on the actual operational structure of the SMK Abdullah Munshi library. Each role is assigned specific responsibilities and access rights to ensure efficient operation, accountability, and data security. The system currently defines three main user roles: Librarian, Student Assistant, and Student (Borrower).

2.1.1 User Role 1: Librarian (Administrator)

User Need

The librarian needs a way to manage the library's book records, borrowing activities, and user accounts efficiently while reducing manual workload and errors.

User Stories

- **LBR-01:** *As a librarian, I want to manage book records so that all library materials are accurately catalogued and searchable.*
- **LBR-02:** *As a librarian, I want to verify and edit OCR-extracted data so that digitized ledger records are accurate before being saved.*
- **LBR-03:** *As a librarian, I want to monitor borrowing and returning activities so that the status of each book is always up to date.*
- **LBR-04:** *As a librarian, I want to manage user accounts and roles so that system access is controlled and secure.*
- **LBR-05:** *As a librarian, I want to generate reports on book inventory and borrowing history so that library usage can be reviewed and audited.*

2.1.2 User Role 2 : Student Assistant

User Need

The student assistant needs a way to assist with borrowing and returning processes efficiently while ensuring that book status updates are accurately recorded in the system.

User Stories

- **STA-01:** *As a student assistant, I want to scan books during borrowing and returning so that the system automatically updates book availability.*

- **STA-02:** As a student assistant, I want the scanner input to be directly recorded in the database so that transactions are fast and error-free.
- **STA-03:** As a student assistant, I want to view borrowing records so that I can assist the librarian in tracking overdue or returned books.
- **STA-04:** As a student assistant, I want to use a simple interface so that daily operations can be carried out without confusion or extensive training.

2.1.3 User Role 3 : Student (Borrower)

User Need

The student needs a way to check book availability, track borrowed books, and receive reminders about due dates.

User Stories

- **STD-01:** *As a student, I want to log in to my account so that my borrowing history can be recorded.*
- **STD-02:** *As a student, I want to view available and borrowed books so that I know which books I can borrow.*
- **STD-03:** *As a student, I want to receive notifications when my borrowed book is nearing its due date so that I can return it on time.*
- **STD-04:** *As a student, I want to book a book online so that I can reserve it before collecting it from the library.*
- **STD-05:** *As a student, I want to favorite a book so that I am notified when it becomes available.*

2.2 System Features

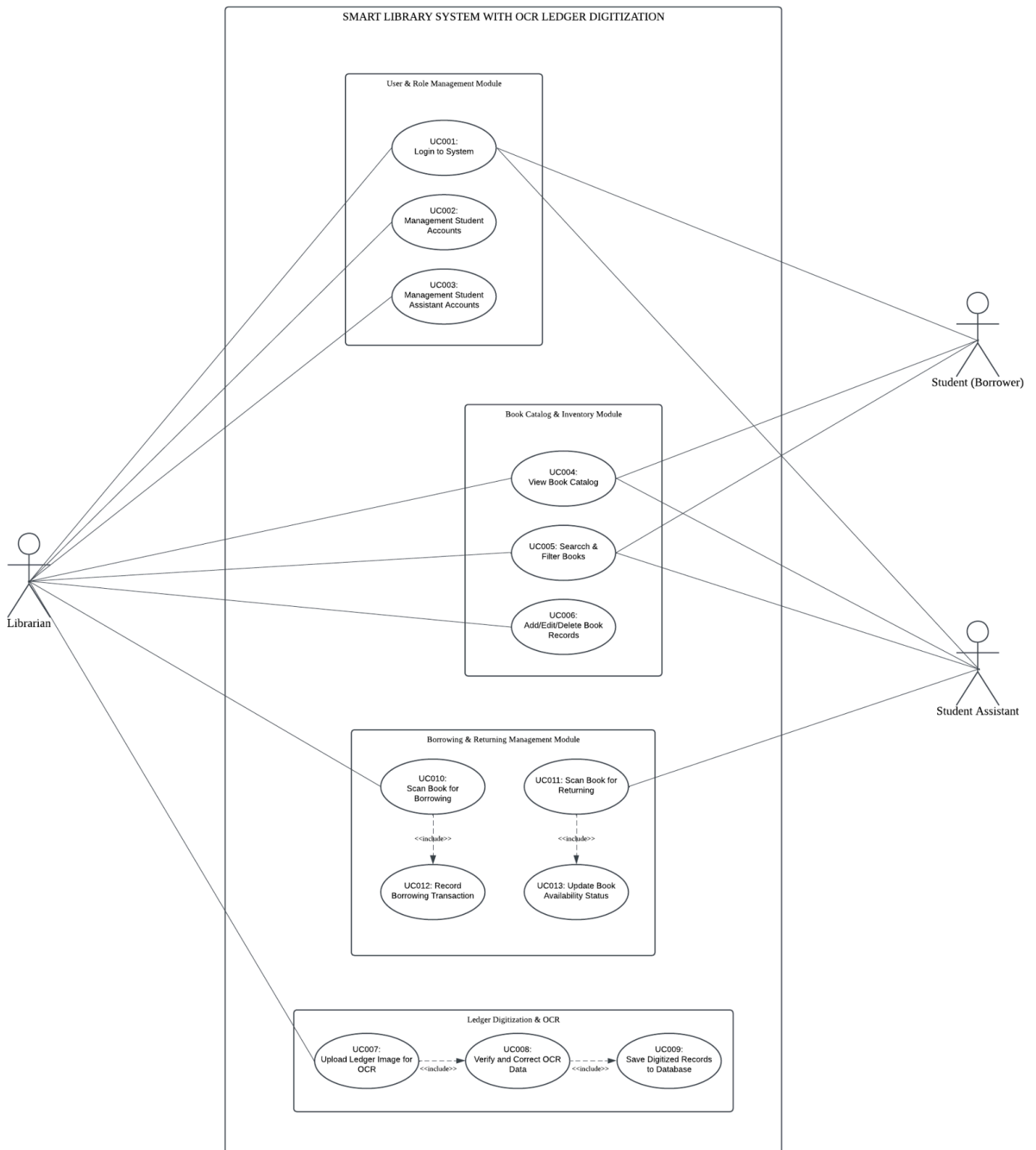


Figure 2.2.1: Use Case Diagram for EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM WITH OCR LEDGER DIGITIZATION

Table 2.1: Description of Module and Functions for ePustaka Munshi: Smart Library System

Use case	Function	Description
UC001	Login to System	This use case allows authorized users (Librarian and Student Assistant) to log in to the system using valid credentials in order to access system functionalities based on their roles.
UC002	Manage Student Accounts	This use case allows the Librarian to create, update, view, and deactivate student user accounts to control access to the library system.
UC003	Manage Student Assistant Accounts	This use case allows the Librarian to manage Student Assistant accounts, including creating, updating, and assigning permissions for assisting library operations.
UC004	View Book Catalog	This use case allows users to view the list of available books in the library, including basic details such as title, author, and availability status.
UC005	Search and Filter Books	This use case allows users to search for books and apply filters (e.g., title, author, category, availability) to quickly locate specific books in the catalog.
UC006	Manage Book Records	This use case allows the Librarian or Student Assistant to add new books, edit existing book details, or delete book records from the system database.
UC007	Upload Ledger Image for OCR	This use case allows the Librarian or Student Assistant to upload scanned images of physical book ledgers for Optical Character Recognition (OCR) processing.
UC008	Verify and Correct OCR Data	This use case allows users to review, validate, and manually correct extracted OCR data before it is saved into the system to ensure data accuracy.
UC009	Save Digitized Records	This use case allows the system to store verified and corrected digitized book records into the database for future access and management.
UC010	Scan Book for Borrowing	This use case allows the Librarian or Student Assistant to scan a book barcode to initiate a borrowing transaction for a student.
UC011	Scan Book for Returning	This use case allows the Librarian or Student Assistant to scan a book barcode to process the return of a borrowed book.
UC012	Validate Book Availability	This use case allows the system to verify whether a book is available for borrowing or already on loan before completing a transaction.
UC013	Update Book Status	This use case allows the system to update the book's availability status and borrowing records after a borrow or return transaction is completed.

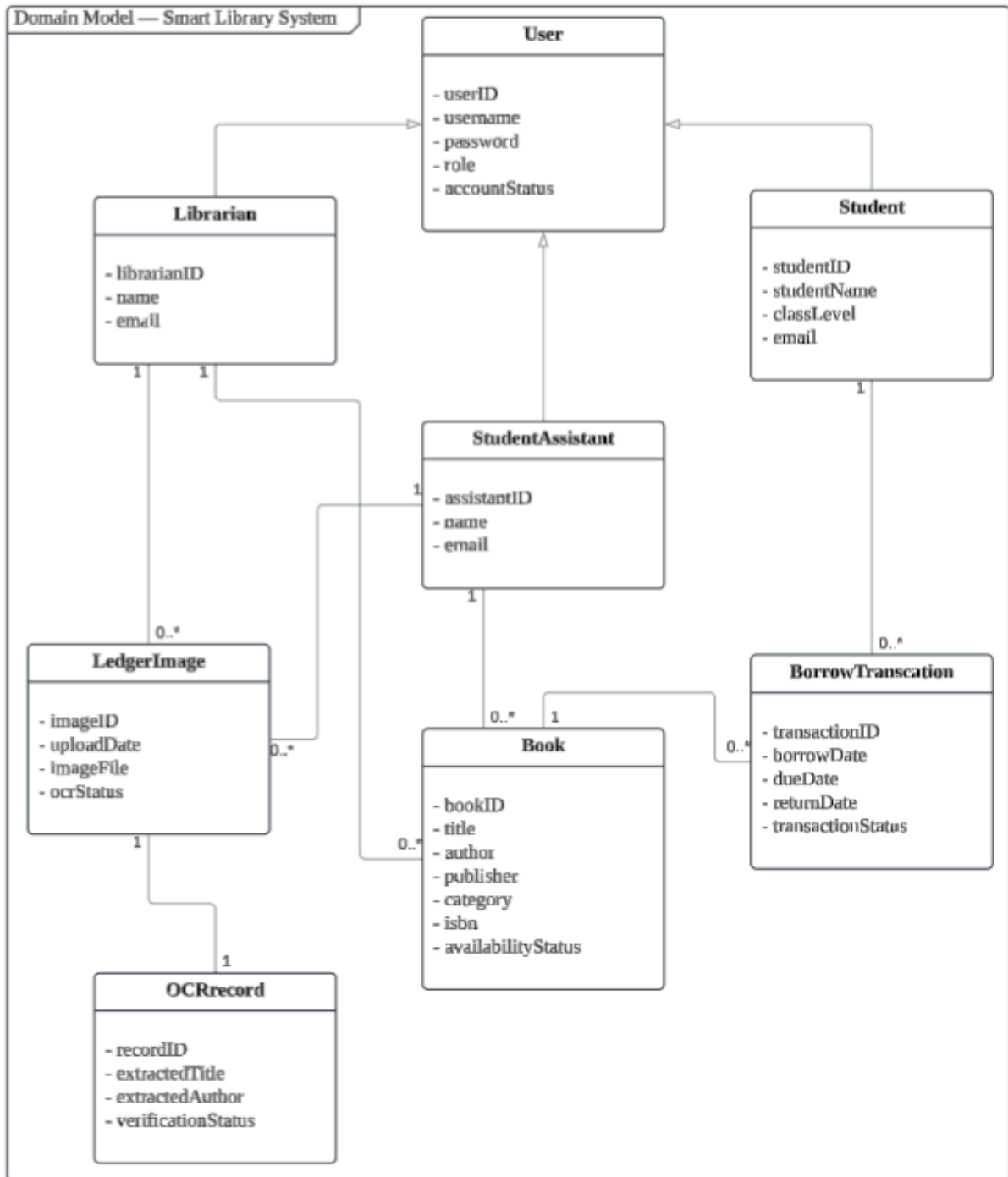


Figure 2.2.2: Domain Model for EPUSTAKA MUNSHI: SMART LIBRARY SYSTEM WITH OCR LEDGER DIGITIZATION

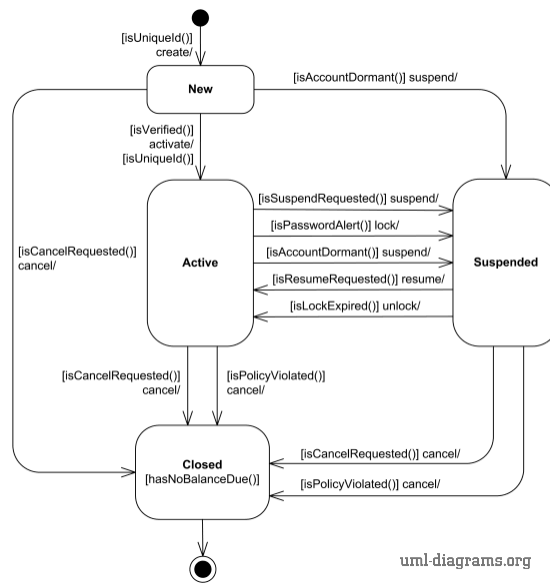


Figure X.X: State Diagram for <Name of the Selected Class>

2.3 Launch Phase

Sprint	ID	User Story	Status	Assignee
Sprint 1	1	US001 – User Login to System	To Do	Wan Zafirzan
	2	US002 – Manage Student Accounts	To Do	Wan Zafirzan
	3	US003 – Manage Student Assistant Accounts	To Do	Wan Zafirzan
Sprint 2	4	US004 – View Book Catalog	To Do	Wan Zafirzan
	5	US005 – Search and Filter Books	To Do	Wan Zafirzan
	6	US006 – Manage Book Records (Add/Edit/Delete)	To Do	Wan Zafirzan
Sprint 3	7	US007 – Upload Ledger Image for OCR	To Do	Wan Zafirzan
	8	US008 – Verify and Correct OCR Data	To Do	Wan Zafirzan
	9	US009 – Save Digitized Book Records	To Do	Wan Zafirzan
Sprint 4	10	US010 – Borrow Book via Barcode Scanning	To Do	Wan Zafirzan

	11	US011 – Return Book via Barcode Scanning	To Do	Wan Zafirzan
	12	US012 – Validate Book Availability	To Do	Wan Zafirzan
	13	US013 – Update Book Status After Transaction	To Do	Wan Zafirzan

2.4 User Story Details

2.4.1 US001: User Story User Login to System

Table 2.1: User Story Description for User Login to System

User Story ID	US001
User Story Name	User Login to System
User Story Description	As a library user (librarian, student assistant, or student), I want to log in to the Smart Library Management System using my registered credentials, So that I can access system functions according to my assigned role.
Acceptance Criteria(s)	<i>Preconditions</i> • User account already exists in the system • User is registered with a valid username and password • System server and database are available <i>Postconditions</i> • User is successfully logged into the system • User session is created • User gains access only to permitted functions based on role <i>Other Conditions</i> • Passwords are stored in encrypted/hashed form • Login session will expire after a predefined period of inactivity
Normal Flow(s)- NF	<i>NF (Main Scenario)</i>

	1. The user opens the Smart Library Management System login page 2. The user enters their username and password 3. The user clicks the “Login” button 4. The system validates the entered credentials 5. The system identifies the user’s role 6. The system redirects the user to the appropriate dashboard
Alternative Flow(s) - AF	<i>AF</i> AF1: Invalid Credentials 1. The user enters an incorrect username or password 2. The system rejects the login attempt 3. The system displays an error message indicating invalid credentials AF2: Inactive Account 1. The user attempts to log in with a disabled account 2. The system denies access 3. The system displays a message indicating account status
Exception Flow(s) - EF	<i>EF</i> EF1: Database Connection Failure 1. The system fails to connect to the authentication database 2. The system displays a system error message 3. The login process is terminated

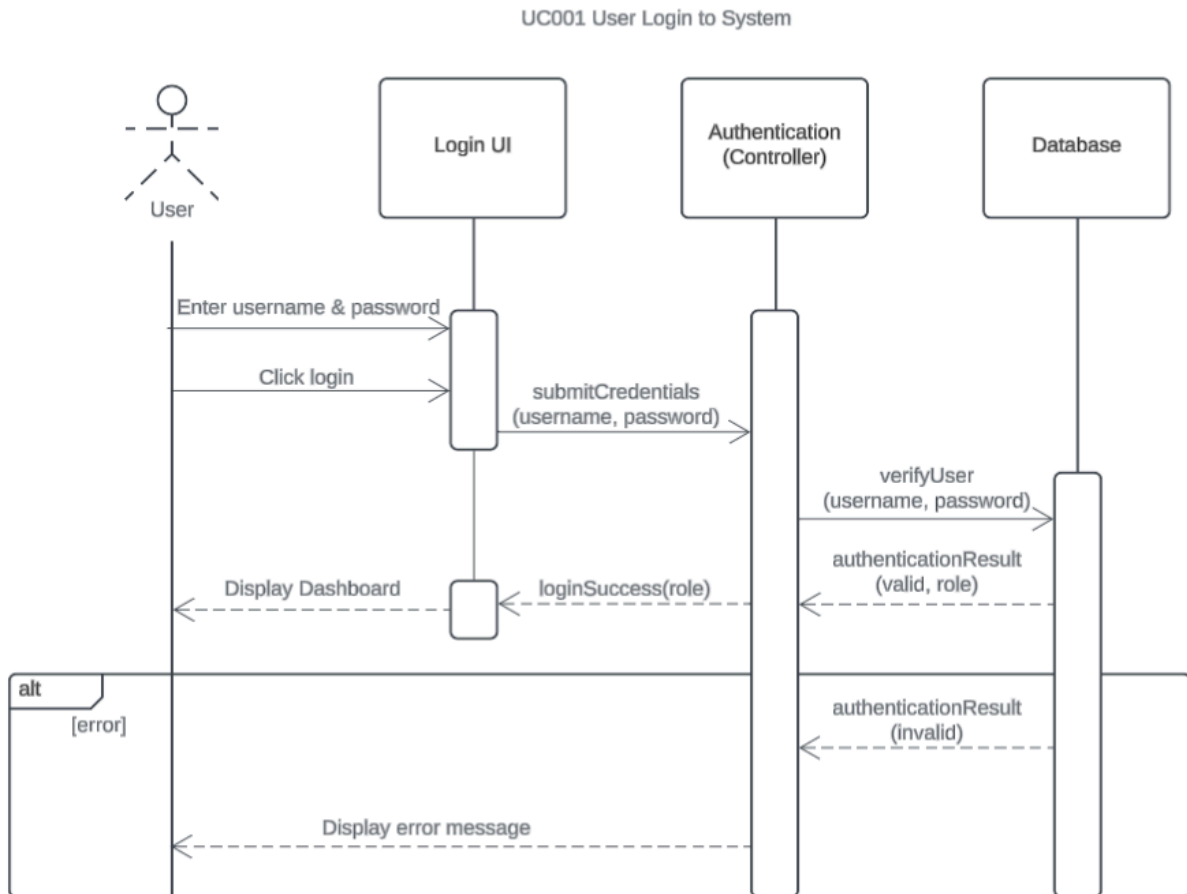


Figure 2.4.1.1: Sequence Diagram for User Login to System

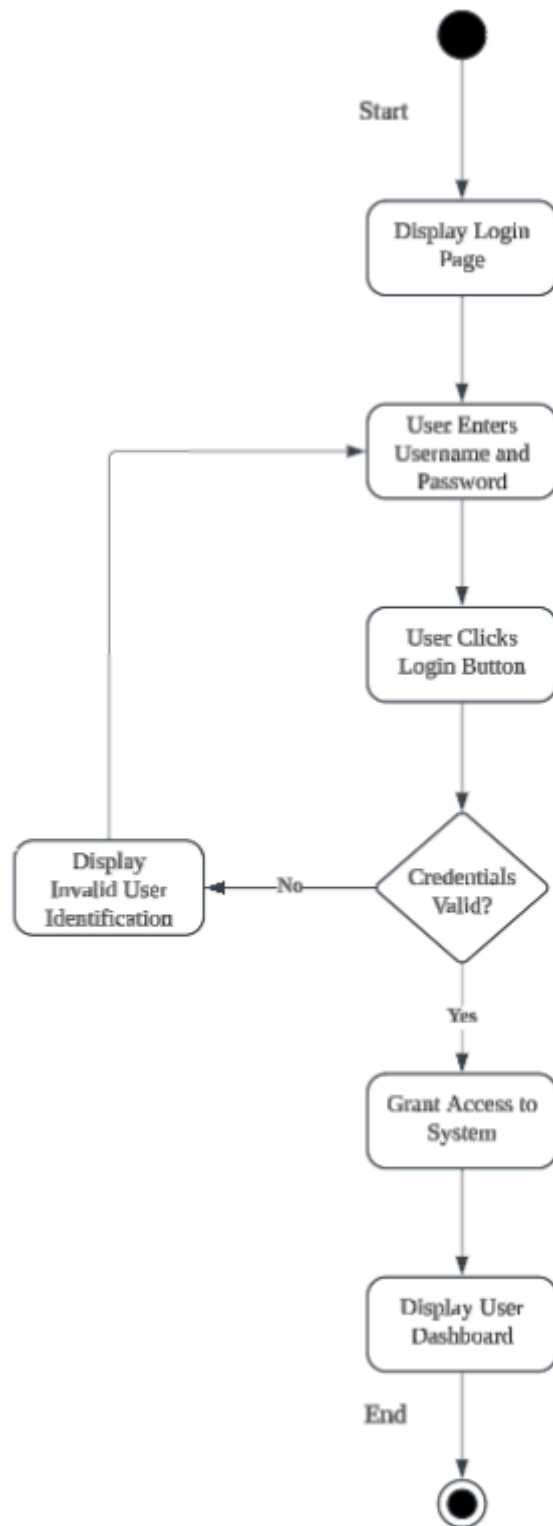


Figure 2.4.1.2: Activity Diagram for User Login to System

2.4.2 US002: User Story Manage Student Accounts

Table 2.2: User Story Description for Manage Student Accounts

User Story ID	US002
User Story Name	Manage Student Accounts
User Story Description	As a librarian, I want to create, update, view, and deactivate student accounts, so that student borrowing activities can be tracked accurately by the system.
Acceptance Criteria(s)	<i>Preconditions</i> • Librarian is logged into the system • Librarian has management access rights <i>Postconditions</i> • Student account information is updated in the system • Student account status reflects the latest changes <i>Other Conditions</i> • Student ID must be unique • Deactivated students cannot borrow books
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The librarian selects “Manage Student Accounts” 2. The system displays a list of student accounts 3. The librarian adds or edits student information 4. The librarian saves the changes 5. The system updates the student account in the database
Alternative Flow(s) - AF	<i>AF</i> AF1: Deactivate Student Account

	1. The librarian selects a student account 2. The librarian chooses “Deactivate” 3. The system updates the account status to inactive
Exception Flow(s) - EF	<i>EF</i> EF1: Duplicate Student ID 1. The librarian enters an existing student ID 2. The system rejects the entry 3. The system displays an error message

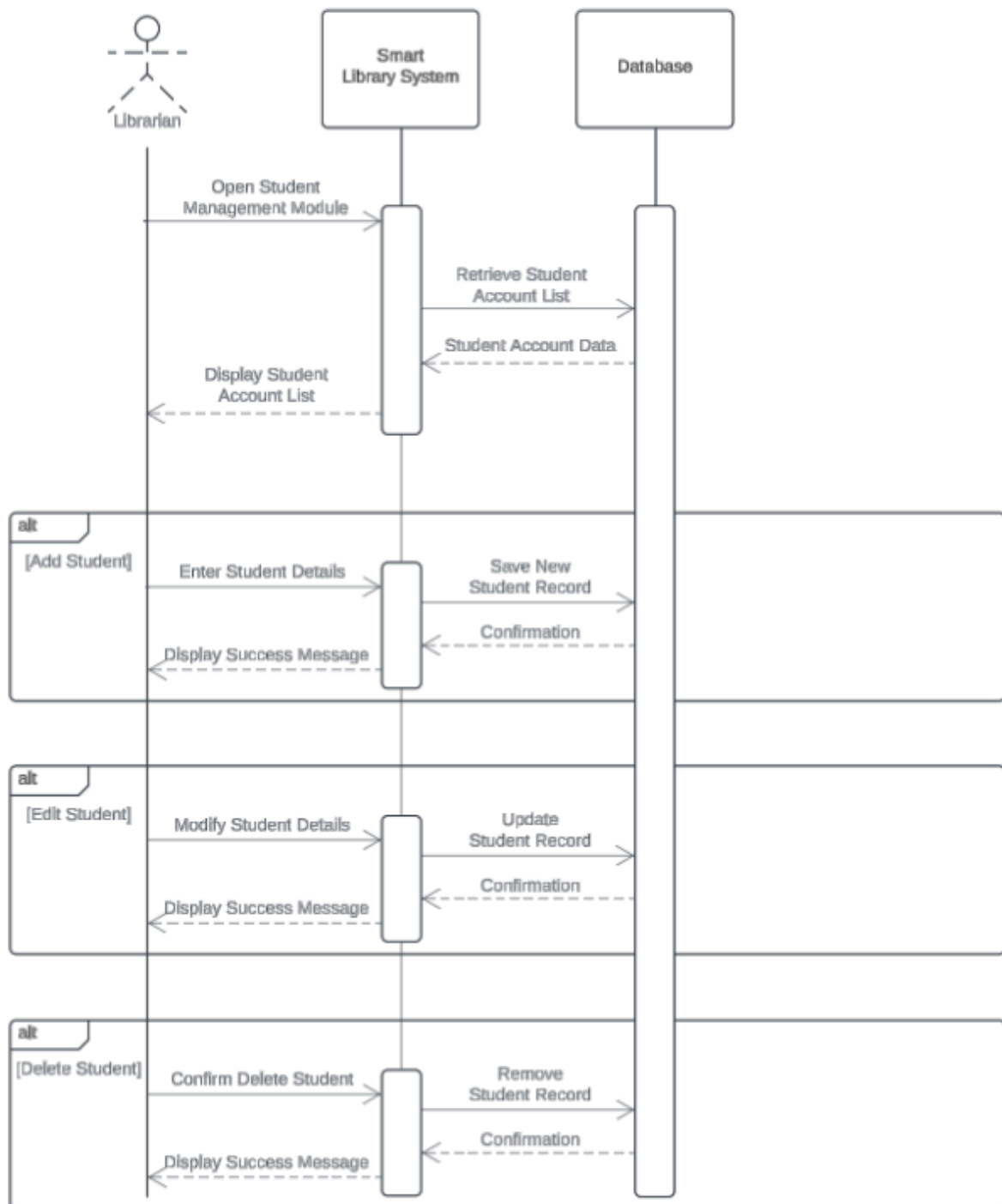


Figure 2.4.2.1: Sequence Diagram for Manage Student Accounts

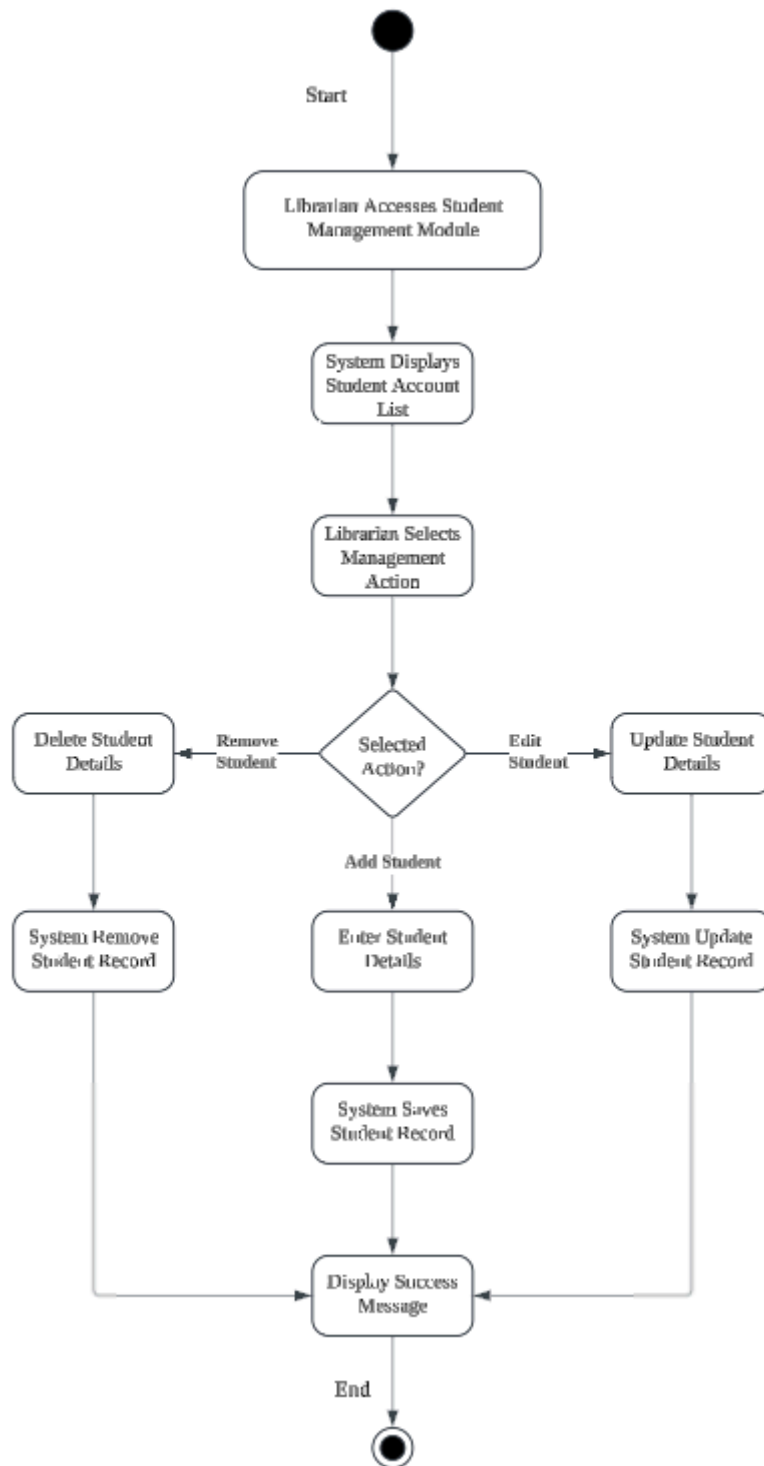


Figure 2.4.2.2: Activity Diagram for Manage Student Accounts

2.4.3 US003: User Story Manage Student Assistant Accounts

Table 2.3: User Story Description for Manage Student Assistant Accounts

User Story ID	US003
User Story Name	Manage Student Assistant Accounts
User Story Description	As a librarian, I want to create, update, view, and deactivate student assistant accounts, so that authorized assistants can help manage book records and borrowing operations.
Acceptance Criteria(s)	<i>Preconditions</i> • Librarian is logged into the system • Librarian has account management privileges <i>Postconditions</i> • Student assistant account is created, updated, or deactivated • Changes are saved in the database <i>Other Conditions</i> • Student assistant username must be unique • Deactivated assistants cannot access the system
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The librarian selects “Manage Student Assistant Accounts” 2. The system displays a list of student assistant accounts 3. The librarian adds or edits assistant details 4. The librarian saves the changes 5. The system updates the account information in the database
Alternative Flow(s) - AF	<i>AF</i> AF1: Deactivate Student Assistant Account

	1. The librarian selects a student assistant account 2. The librarian chooses “Deactivate” 3. The system updates the account status to inactive
Exception Flow(s) - EF	<i>EF</i> EF1: Duplicate Username 1. The librarian enters an existing username 2. The system rejects the entry 3. The system displays an error message

UC003 Manage Student Assistant Accounts

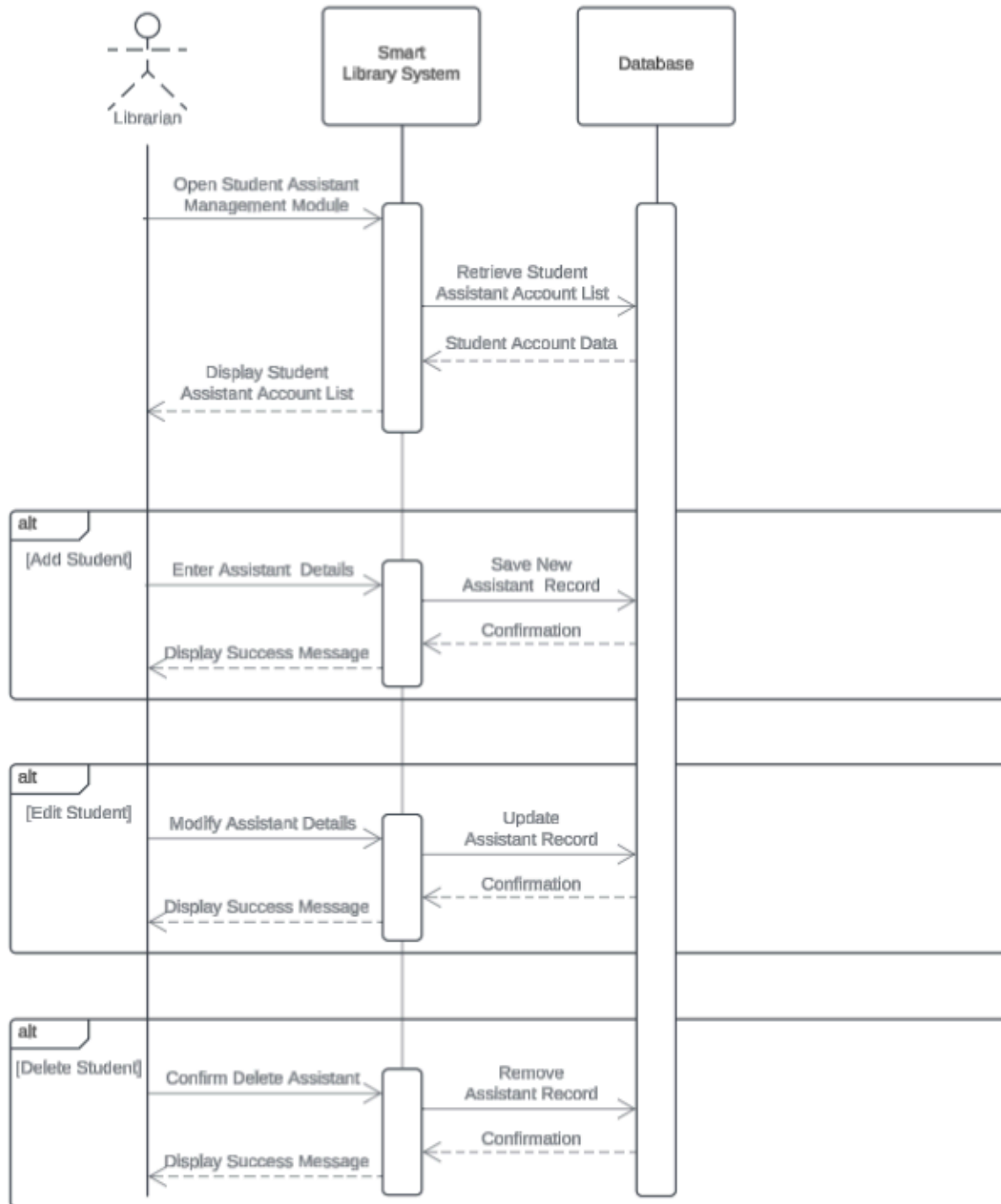


Figure 2.4.3.1: Sequence Diagram for Manage Student Assistant Accounts

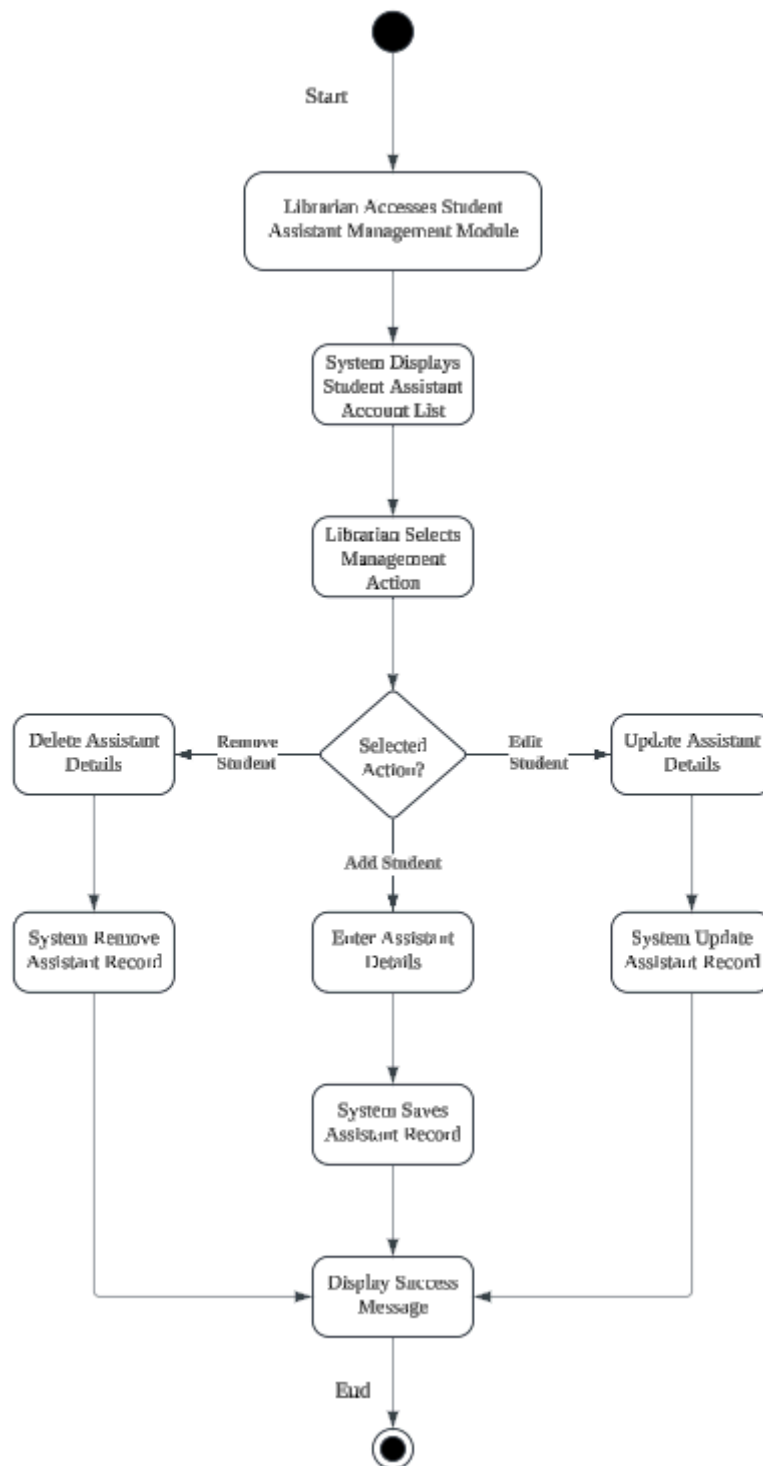


Figure 2.4.3.2: Activity Diagram for Manage Student Assistant Accounts

2.4.4 US004 User Story View Book Catalog

Table 2.4: User Story Description for View Book Catalog

User Story ID	US004
User Story Name	View Book Catalog
User Story Description	As a student, student assistant, or librarian, I want to view the library's book catalog, so that I can know what books exist and check their availability status.
Acceptance Criteria(s)	<i>Preconditions</i> • User is logged into the system <i>Postconditions</i> • Book catalog is displayed to the user <i>Other Conditions</i> • Only active books are shown in the catalog • Availability status is updated in real time
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The user selects "View Book Catalog" 2. The system retrieves book records from the database 3. The system displays the book list with availability status
Alternative Flow(s) - AF	<i>AF</i> AF1: Empty Catalog 1. The system finds no book records 2. The system displays a "No books available" message
Exception Flow(s) - EF	<i>EF</i>

EF1: Database Connection Error

1. The system fails to retrieve book data
2. The system displays an error message

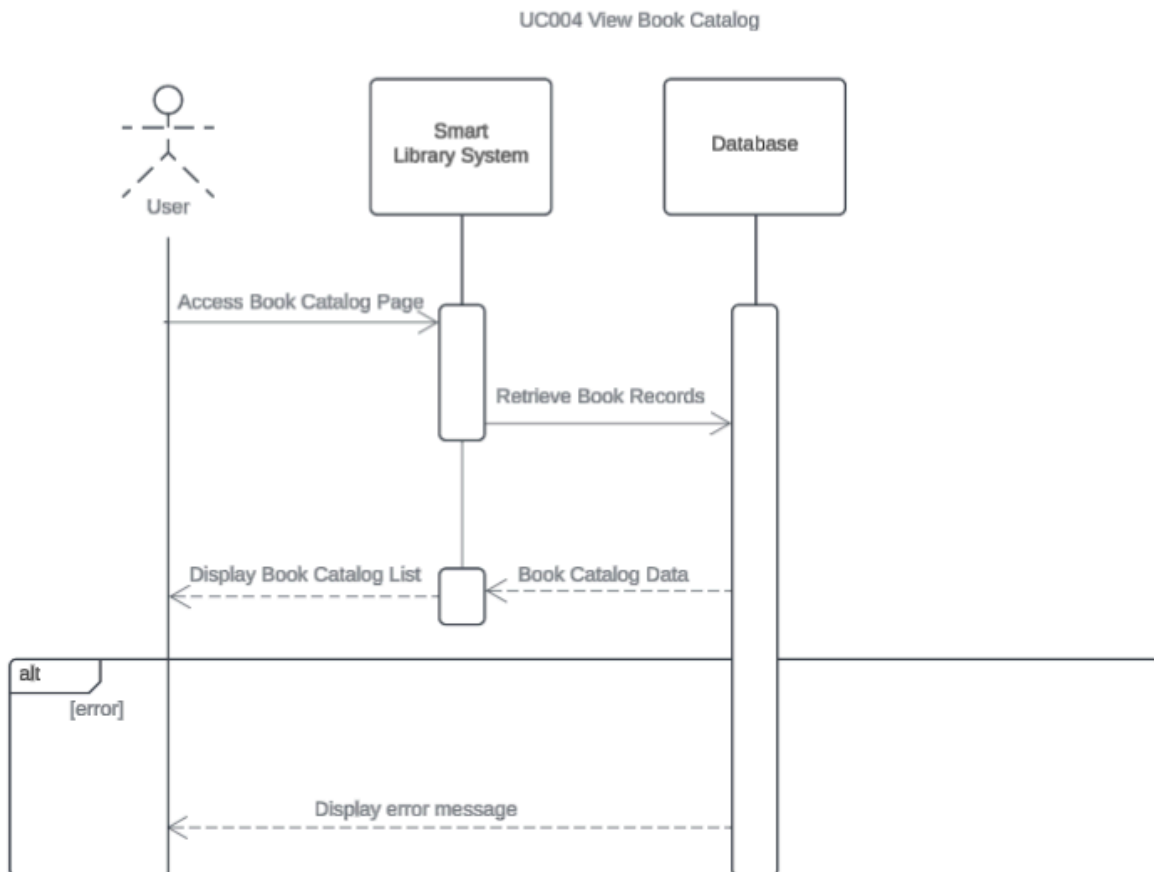


Figure 2.4.4.1: Sequence Diagram for View Book Catalog

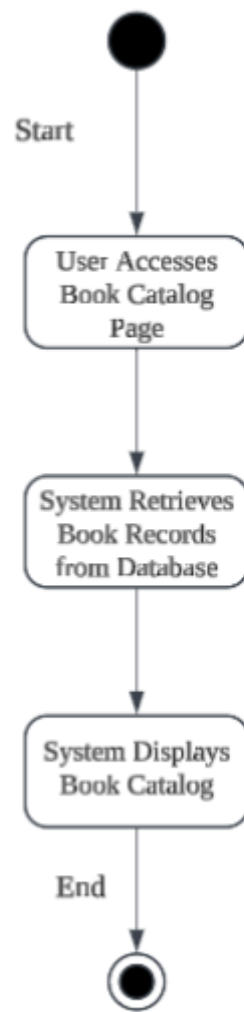


Figure 2.4.4.2: Activity Diagram for View Book Catalog

2.4.5 US005 User Story Search & Filter Books

Table 2.5: User Story Description for Search & Filter Books

User Story ID	US005
User Story Name	Search & Filter Books
User Story Description	As a student, student assistant, or librarian, I want to search and filter books in the catalog, so that I can quickly find specific books based on keywords or categories.
Acceptance Criteria(s)	<i>Preconditions</i> • User is logged into the system • Book catalog exists in the database <i>Postconditions</i> • Filtered or searched book list is displayed <i>Other Conditions</i> • Search is case-insensitive • Multiple filters can be applied together
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The user enters a keyword or selects filter options 2. The system processes the search or filter criteria 3. The system retrieves matching books from the database 4. The system displays the filtered book list
Alternative Flow(s) - AF	<i>AF</i> AF1: No Matching Results 1. The system finds no books matching the criteria 2. The system displays a “No results found” message

**Exception
Flow(s) - EF**

EF

EF1: System Search Failure

1. The system encounters an error during search processing
2. The system displays an error message

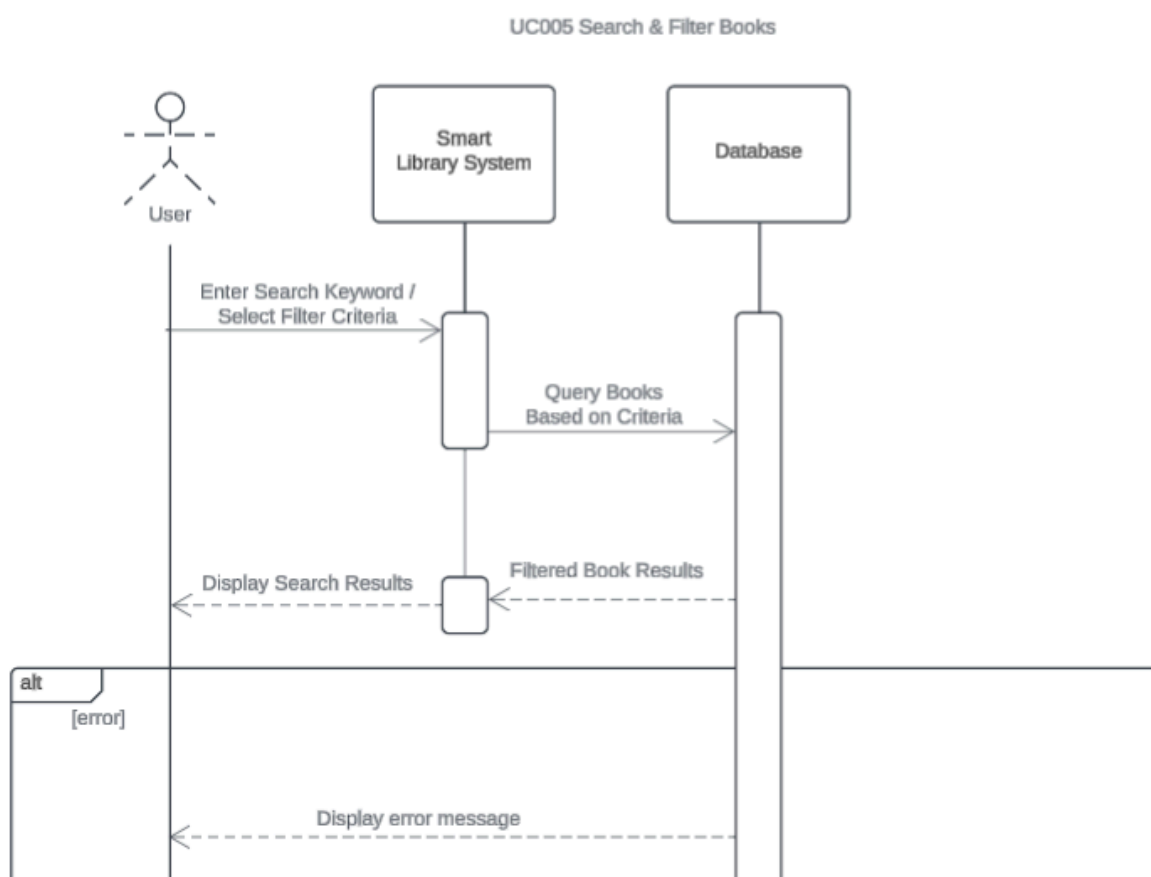


Figure 2.4.5.1: Sequence Diagram for Search & Filter Books

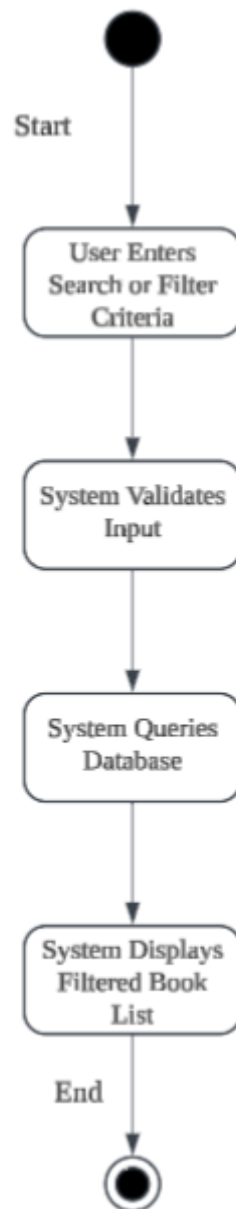


Figure 2.4.5.2: Activity Diagram for View Book Catalog

2.4.6 US006 User Story Add / Edit / Delete Book Records

Table 2.6: User Story Description for Add / Edit / Delete Book Records

User Story ID	US006
User Story Name	Add / Edit / Delete Book Records
User Story Description	As a librarian, I want to add, edit, or delete book records, so that the book catalog remains accurate and up to date.

Acceptance Criteria(s)	<i>Preconditions</i> • Librarian is logged into the system • Librarian has management access rights <i>Postconditions</i> • Book record is added, updated, or removed in the database <i>Other Conditions</i> • Mandatory fields must not be empty • Deleted records are permanently removed
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The librarian selects add, edit, or delete book option 2. The librarian enters or updates book information 3. The system validates the input data 4. The system saves the changes to the database 5. The system displays a confirmation message
Alternative Flow(s) - AF	<i>AF</i> AF1: Cancel Operation 1. The librarian cancels the operation 2. The system returns to the book management page
Exception Flow(s) - EF	<i>EF</i> EF1: Data Validation Error 1. The system detects missing or invalid data 2. The system displays an error message

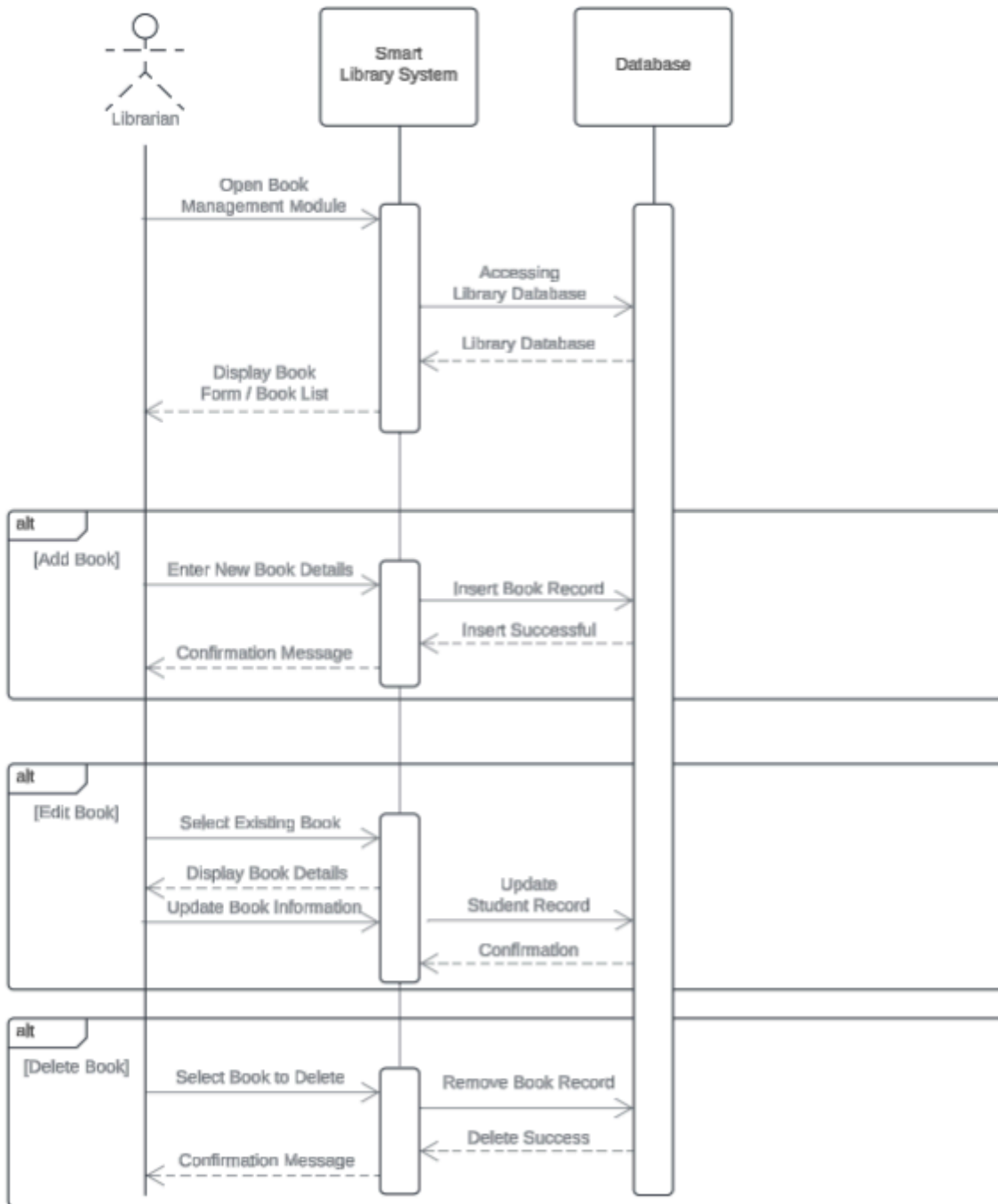


Figure 2.4.6.1: Sequence Diagram for Add / Edit / Delete Book Records

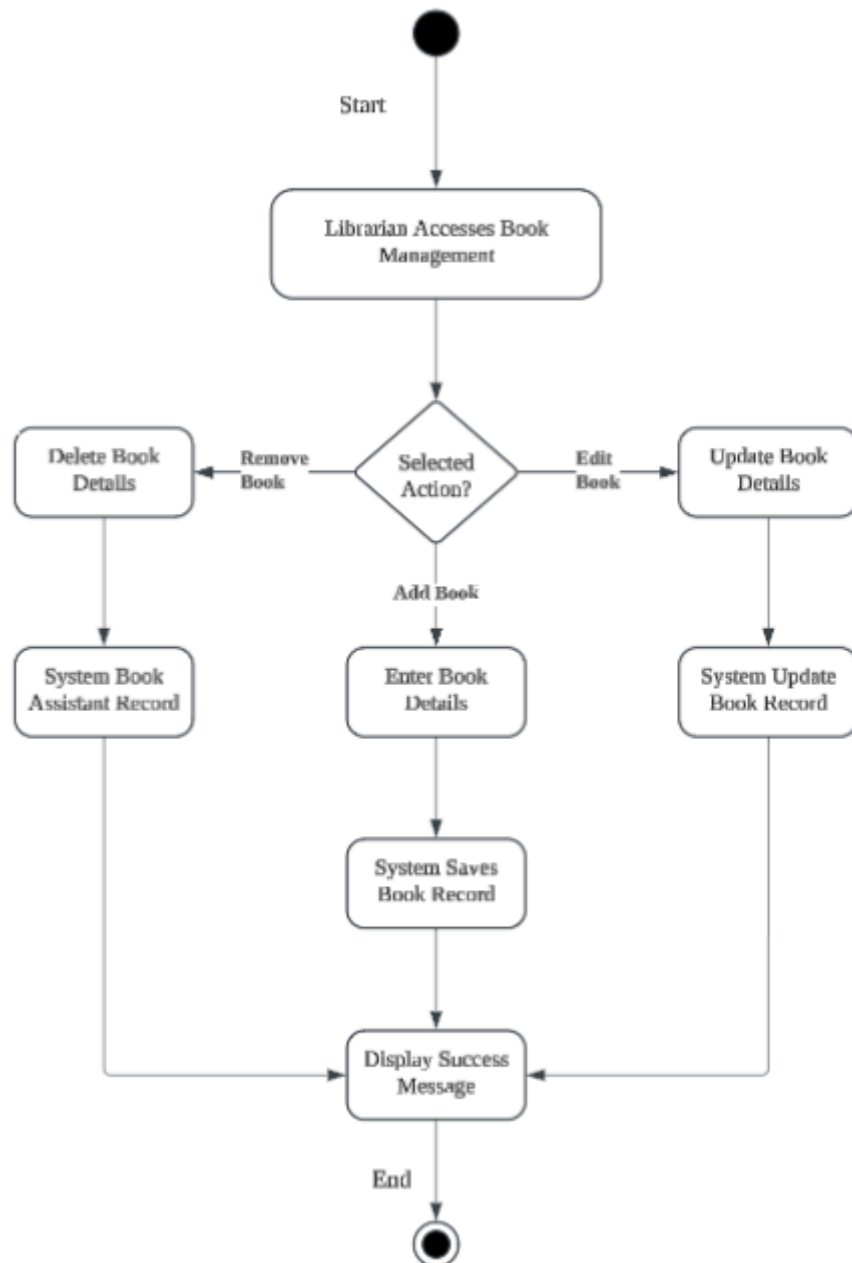


Figure 2.4.6.2: Activity Diagram for Add / Edit / Delete Book Records

2.4.7 US007 User Story Upload Ledger Image for OCR

Table 2.7: User Story Description for Upload Ledger Image for OCR

User Story ID	US007
User Story Name	Upload Ledger Image for OCR
User Story Description	As a librarian, I want to upload images of the handwritten ledger, so that the system can extract book data using OCR to digitize the records.
Acceptance Criteria(s)	<i>Preconditions</i> • Librarian is logged in with management access • Ledger images are available in digital format <i>Postconditions</i> • Ledger images are saved in the system • OCR module is ready to process the uploaded images <i>Other Conditions</i> • Images must be legible for accurate OCR extraction • System may store multiple ledger images per upload session
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. Librarian selects the ledger upload option 2. Librarian chooses one or more ledger images from the computer 3. System uploads and stores the images 4. System confirms successful upload
Alternative Flow(s) - AF	<i>AF</i> AF1: Cancel Upload 1. Librarian cancels the upload process

	2. System returns to the previous screen without saving images
Exception Flow(s) - EF	EF EF1: Invalid File Format 1. System detects unsupported file type 2. System displays an error message and prompts for a valid format EF2: Upload Failure 1. System fails to store the image due to network or storage issue 2. System displays an error and allows retry

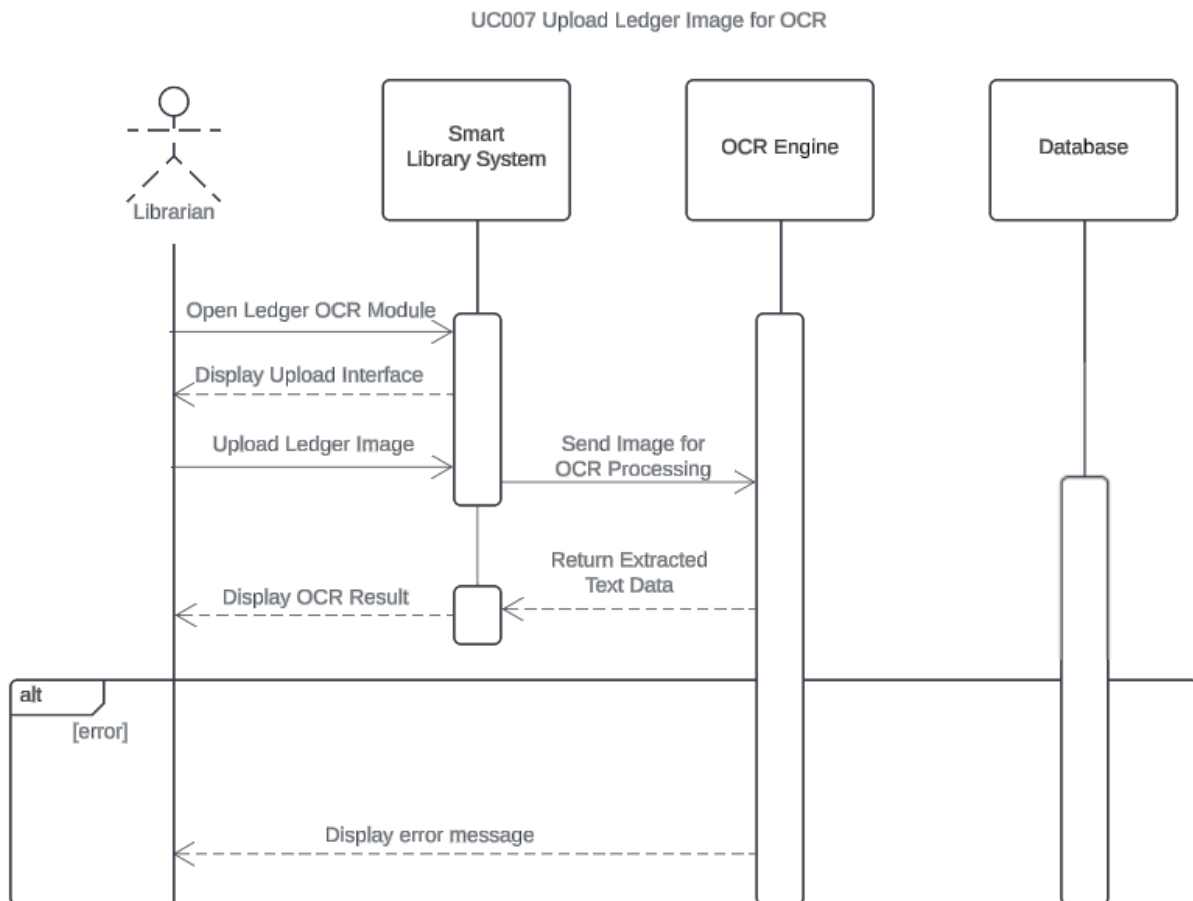


Figure 2.4.7.1: Sequence Diagram for Upload Ledger Image for OCR



Figure 2.4.7.2: Activity Diagram for Upload Ledger Image for OCR

2.4.8 US008 User Story Verify and Correct OCR Data

Table 2.8: User Story Description for Verify and Correct OCR Data

User Story ID	US008
User Story Name	Verify and Correct OCR Data
User Story Description	As a librarian, I want to review and correct data extracted by the OCR system, so that the digitized book records are accurate and ready to be saved in the database.
Acceptance Criteria(s)	<i>Preconditions</i> • OCR processing of the ledger image is completed • Librarian is logged into the system with appropriate access <i>Postconditions</i> • Corrected data is ready to be stored in the database • Any changes made by the librarian are reflected in the verified dataset <i>Other Conditions</i> • System should highlight likely OCR errors for easier review • Changes must be tracked for audit purposes
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The system displays the OCR-extracted data to the librarian 2. The librarian reviews each record for accuracy 3. The librarian corrects errors or fills in missing data 4. The librarian confirms the review 5. The system marks the data as verified and ready for saving
Alternative Flow(s) - AF	<i>AF</i> AF1: Skip a Record

	1. The librarian decides to skip a particular record for later review 2. The system saves the unverified record in a pending state
Exception Flow(s) - EF	<i>EF</i> EF1: OCR Data Missing Entirely 1. The system detects no data extracted from the uploaded image 2. The system notifies the librarian to re-upload the ledger image

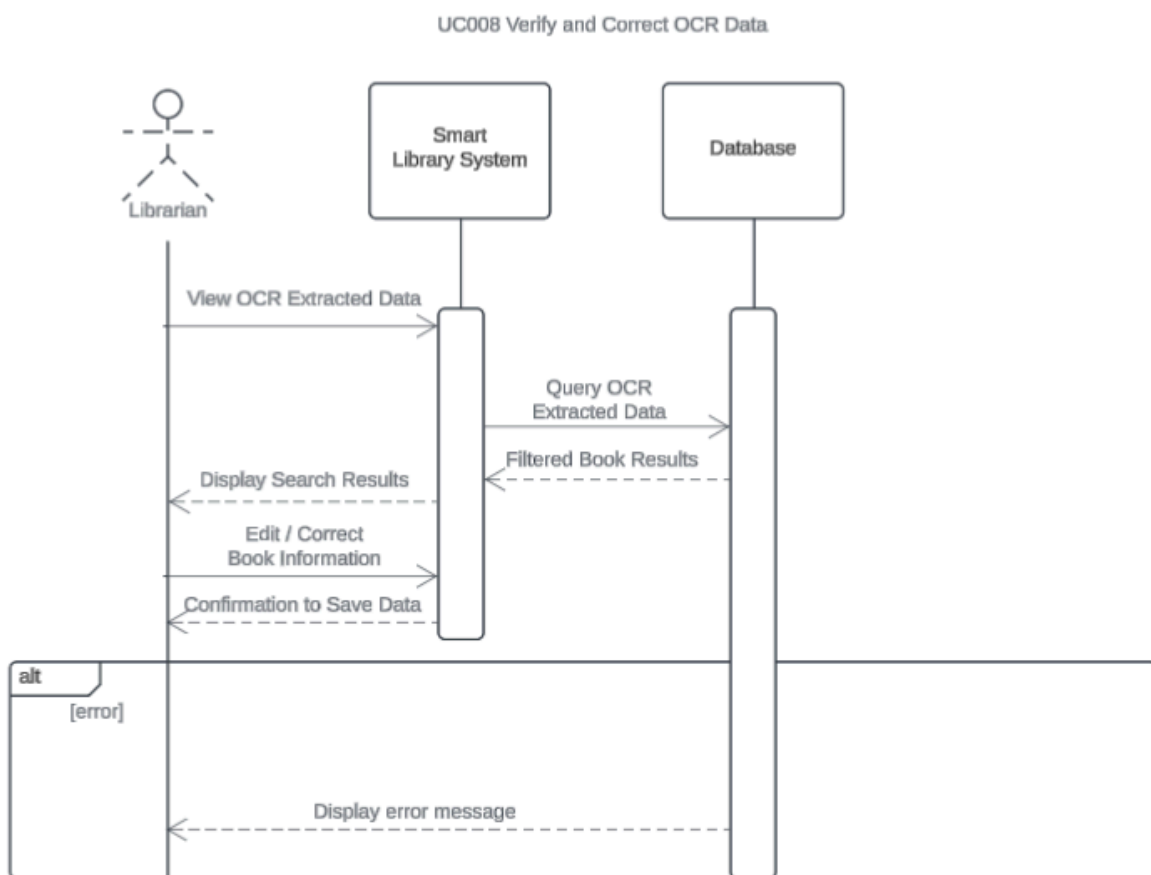


Figure 2.4.8.1: Sequence Diagram for Verify and Correct OCR Data

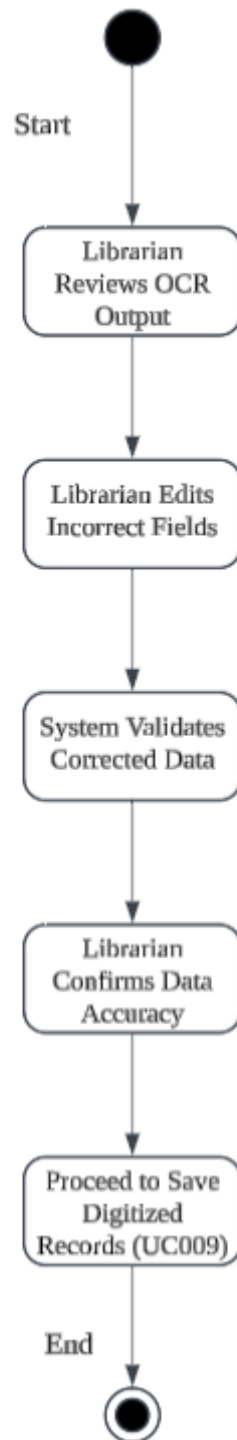


Figure 2.4.8.2: Activity Diagram for Verify and Correct OCR Data

2.4.9 US009 User Story Save Digitized Records to Database

Table 2.9: User Story Description for Save Digitized Records to Database

User Story ID	US009
User Story Name	Save Digitized Records to Database
User Story Description	As a librarian, I want to save the verified and corrected OCR data into the system database, so that all historical book records are preserved digitally and available for future library operations.
Acceptance Criteria(s)	<i>Preconditions</i> • OCR data has been verified and corrected by the librarian • Librarian is logged in with permission to modify the database <i>Postconditions</i> • Digitized book records are stored in the system database • Data is available for other modules (search, borrow/return, reporting) <i>Other Conditions</i> • System should prevent duplicate entries when saving • System should notify the librarian if saving fails
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The librarian clicks “Save” on the verified OCR dataset 2. The system validates the data for completeness and duplicates 3. The system stores the data in the database 4. The system confirms successful saving to the librarian
Alternative Flow(s) - AF	<i>AF</i> AF1: Save Partially Verified Data

	1. The librarian attempts to save only some verified records 2. The system stores only the verified records and keeps unverified ones pending
Exception Flow(s) - EF	<i>EF</i> EF1: Database Connection Failure 1. The system fails to connect to the database 2. The system notifies the librarian and prompts to retry

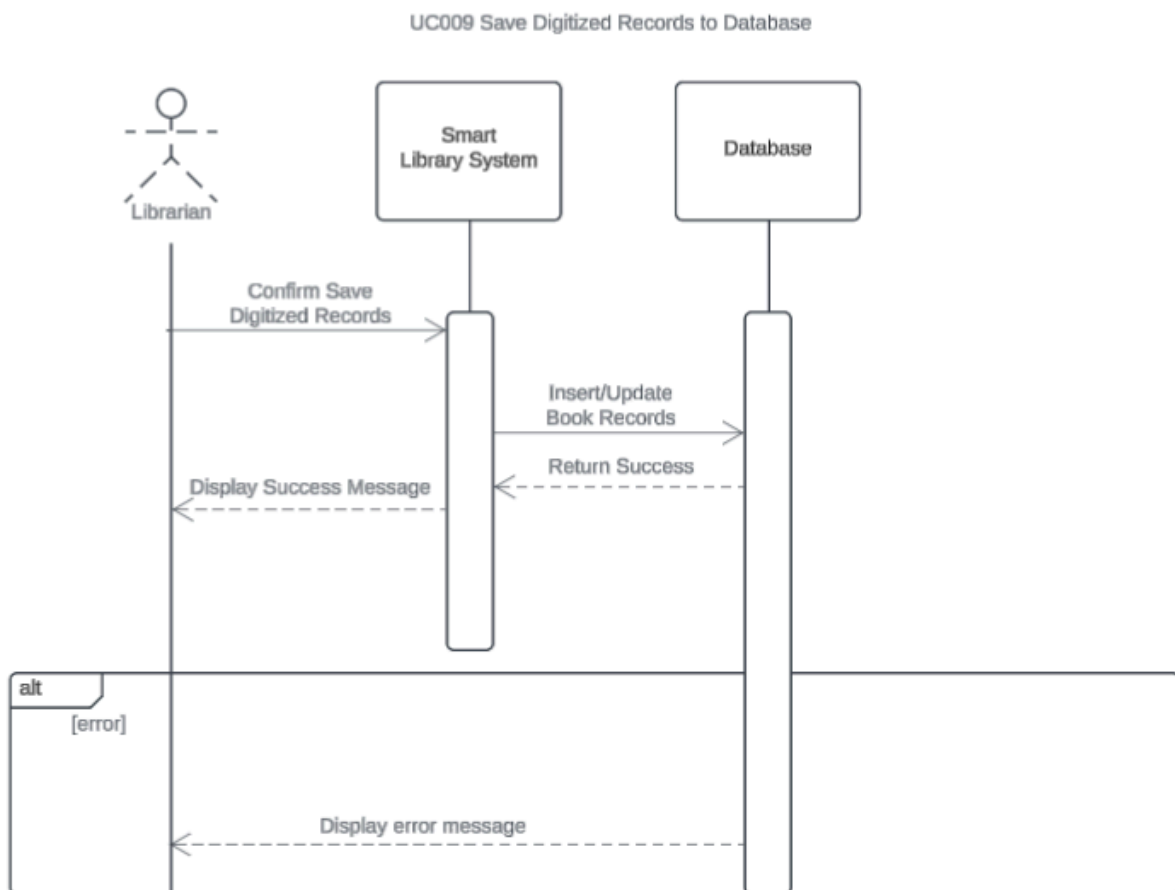


Figure 2.4.9.1: Sequence Diagram for Save Digitized Records to Database



Figure 2.4.9.2: Activity Diagram for Save Digitized Records to Database

2.4.10 US010 User Story Scan Book for Borrowing

Table 2.10: User Story Description for Scan Book for Borrowing

User Story ID	US010
User Story Name	Scan Book for Borrowing
User Story Description	As a student assistant, I want to scan a book using the barcode scanner for borrowing, so that the system can automatically record that the book is being borrowed and update its status.
Acceptance Criteria(s)	<i>Preconditions</i> • Student assistant is logged in • Book is physically present and has a valid barcode/QR code • Scanner is connected to the system <i>Postconditions</i> • Book's status is updated as borrowed • Borrowing transaction is recorded in the database • Borrower can see the book in their account <i>Other Conditions</i> • If the book is already borrowed, the system should notify the assistant • System should log the borrowing activity for reporting purposes
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. The student assistant selects "Borrow Book" in the system 2. The assistant scans the book's barcode/QR code 3. The system verifies book availability 4. The system records the borrowing transaction

	5. The system updates the book's status as borrowed 6. The system confirms successful borrowing to the assistant
Alternative Flow(s) - AF	<i>AF</i> AF1: Book Already Borrowed 1. The assistant scans a book that is currently borrowed 2. The system displays a warning and prevents the borrowing
Exception Flow(s) - EF	<i>EF</i> EF1: Scanner Malfunction 1. The scanner fails to read the barcode/QR code 2. The system prompts the assistant to retry or manually enter the book ID

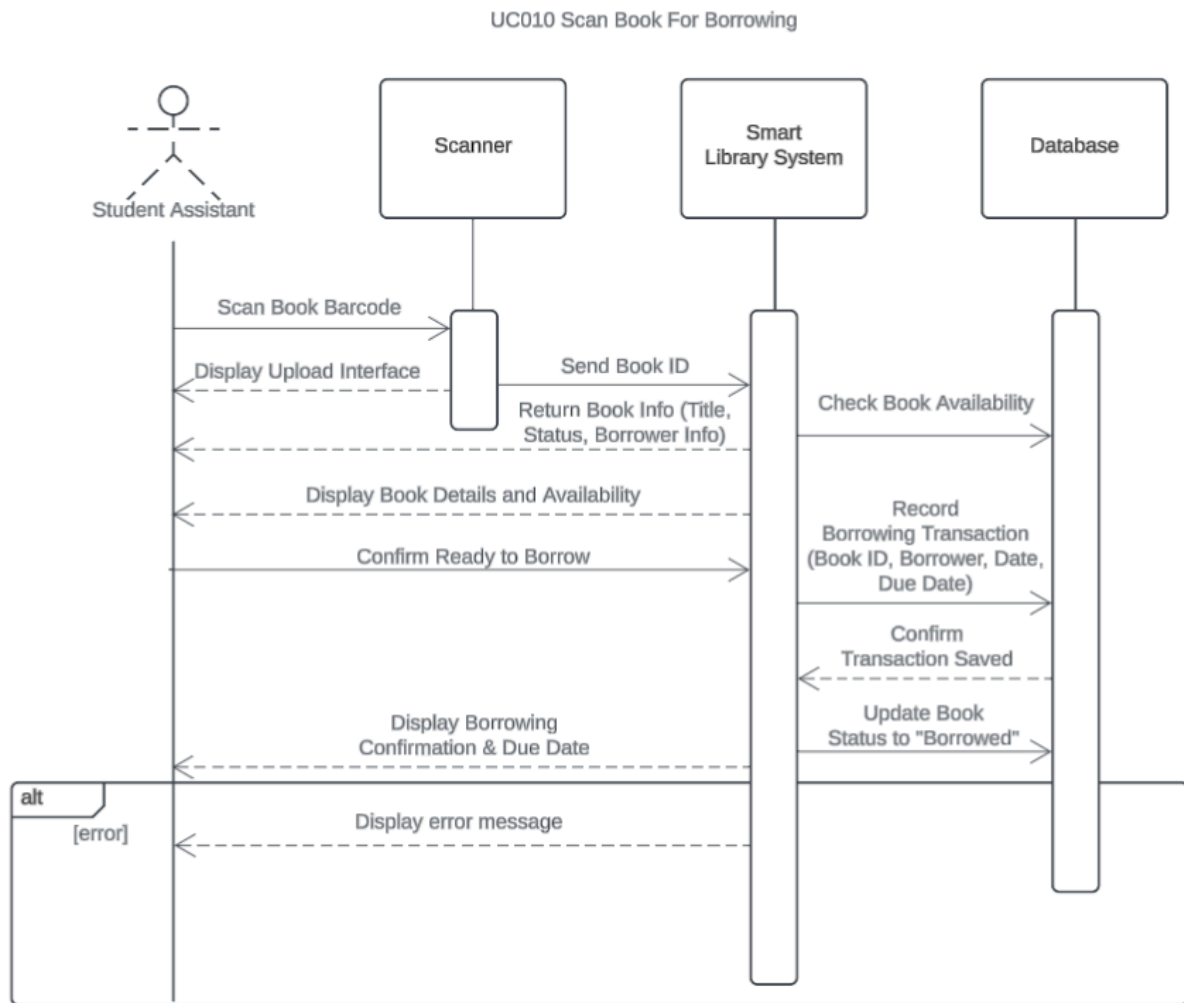
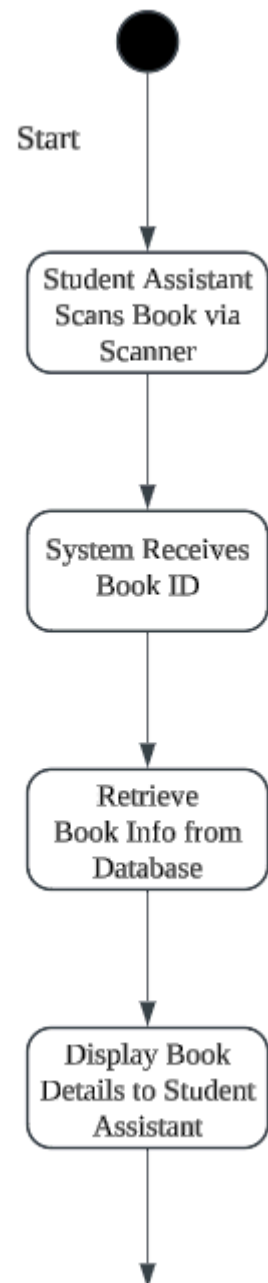


Figure 2.4.10.1: Sequence Diagram for Scan Book For Borrowing



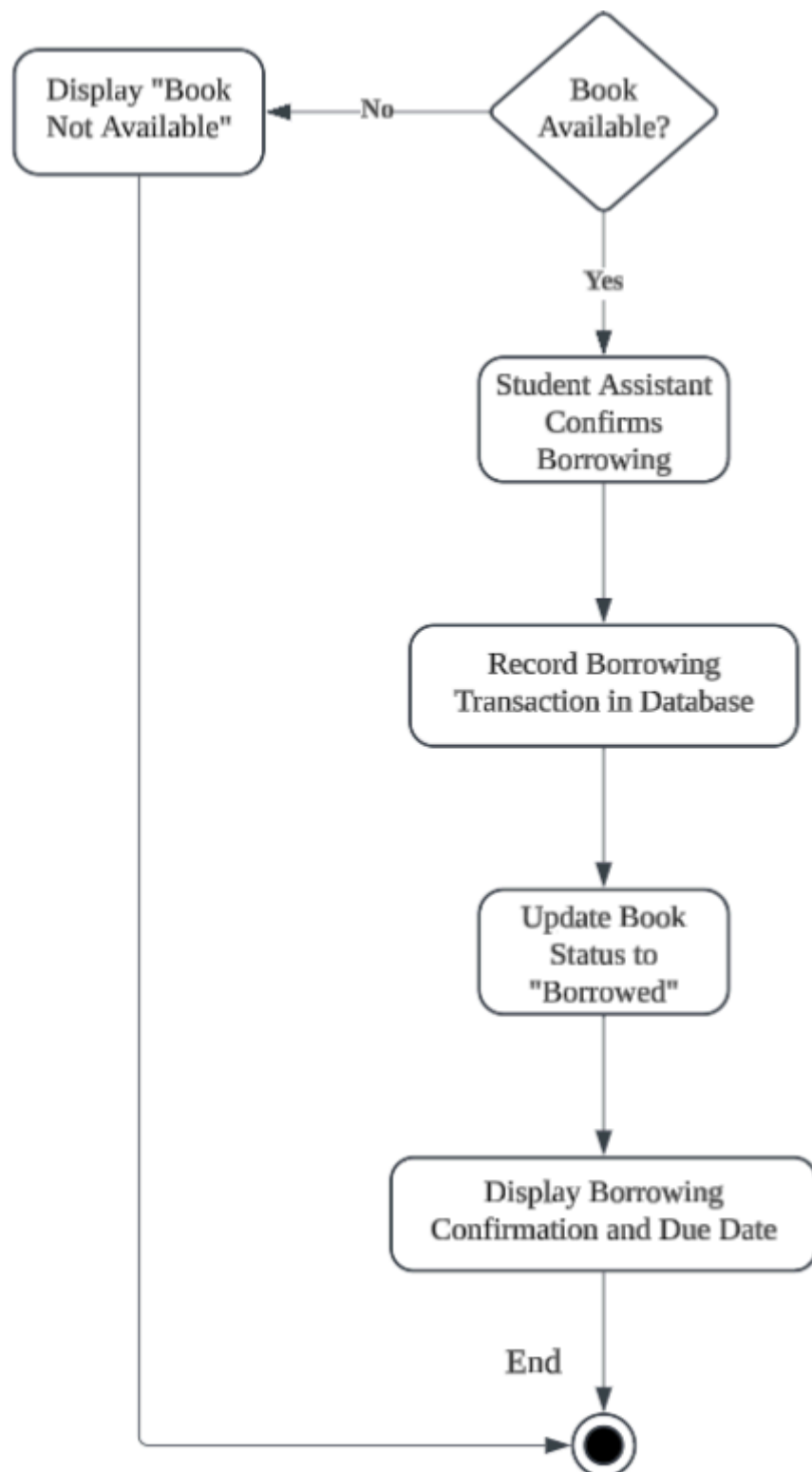


Figure 2.4.10.2: Activity Diagram for Scan Book For Borrowing

2.4.11 US011 User Story Scan Book for Returning

Table 2.11: User Story Description for User Story Description for Scan Book for Returning

User Story ID	US011
User Story Name	User Story Description for Scan Book for Returning
User Story Description	As a student assistant, I want to scan books when they are returned, so that the system updates the book's status and ensures it is available for borrowing again.
Acceptance Criteria(s)	<i>Preconditions</i> • Student assistant is logged in • Book has been borrowed previously (exists in the borrowing records) <i>Postconditions</i> • Book status changes from “borrowed” to “available” • Borrower record updated with the return date • System is ready for next borrowing transaction <i>Other Conditions</i> • If the book was not recorded as borrowed, the system alerts the assistant • Late returns can trigger overdue notifications
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 1. Student assistant scans the returned book 2. System retrieves borrowing transaction for that book 3. System updates the book status to “available” 4. System records the return date in the borrower's account 5. System confirms to the student assistant that the return is successfully processed

Alternative Flow(s) - AF	AF AF1: Book Not Found in Borrowed Records 1. System cannot find a matching borrowing record for the scanned book 2. System alerts the assistant to manually check and correct the record
Exception Flow(s) - EF	EF EF1: Scanner or Database Error 1. System fails to read the book code or update the database 2. System displays an error and requests the assistant to retry

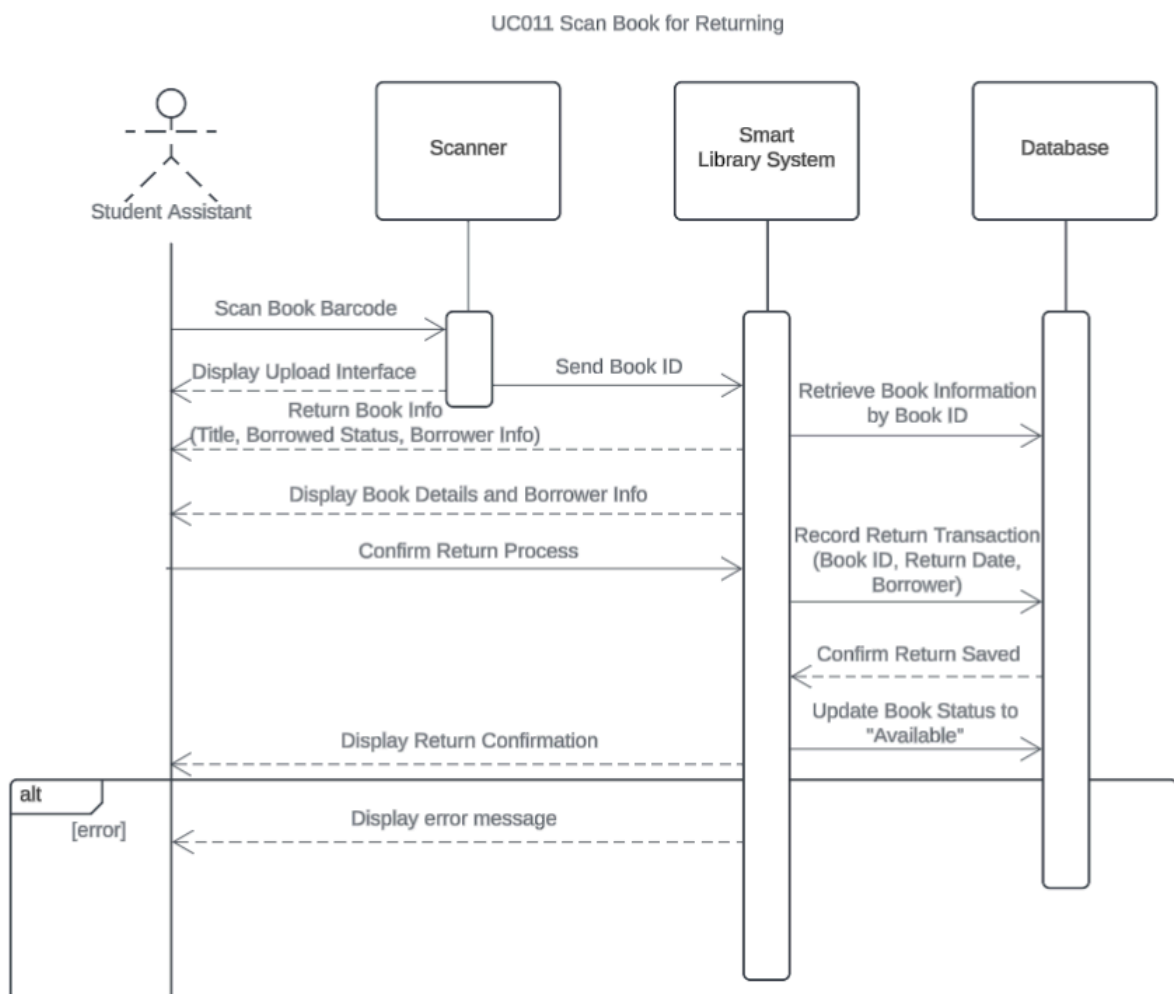
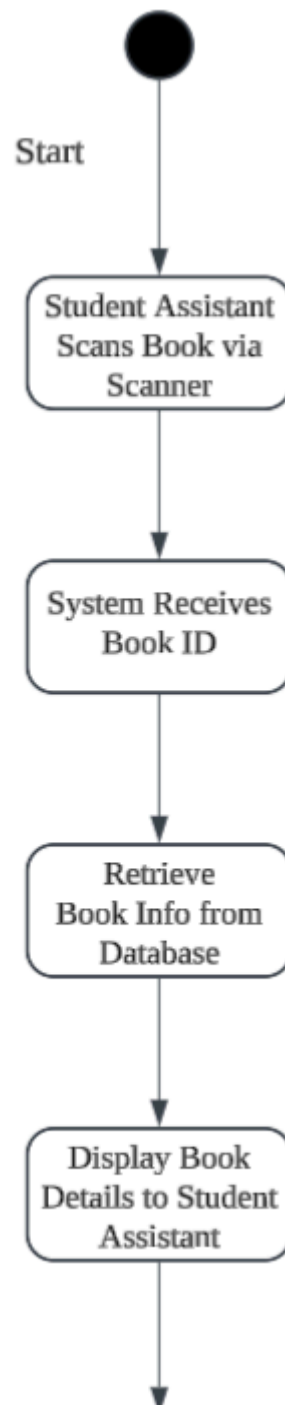


Figure 2.4.11.1: Sequence Diagram for Scan Book for Returning



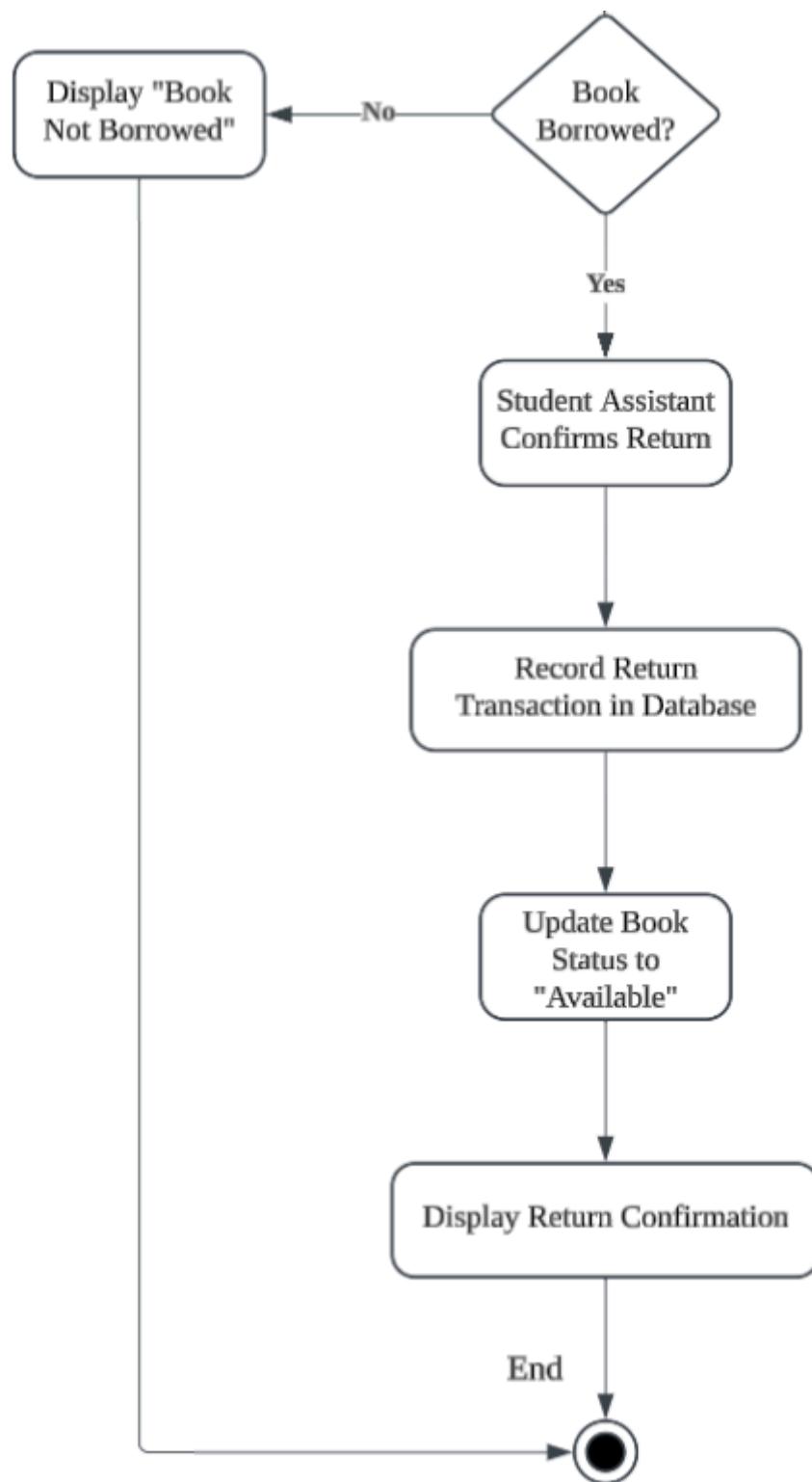


Figure 2.4.11.2: Activity Diagram for Scan Book for Returning

2.4.12 US012 User Story Record Borrowing Transaction

Table 2.12: User Story Description for Record Borrowing Transaction

User Story ID	US012
User Story Name	Record Borrowing Transaction
User Story Description	As a student assistant, I want the system to automatically record borrowing transactions after scanning a book, so that the library keeps accurate records of which student has borrowed which book and when it is due for return.
Acceptance Criteria(s)	<i>Preconditions</i> • Student assistant is logged in • Book has been scanned and verified for availability • Borrower's student account exists in the system <i>Postconditions</i> • Borrowing transaction is successfully saved • Book status updated to "borrowed" • Borrower account shows the borrowed book and due date <i>Other Conditions</i> • If the book cannot be recorded due to a database error, the system should alert the student assistant • Borrowing history should be retained for audit purposes
Normal Flow(s)- NF	<i>NF (Main Scenario)</i> 7. The student assistant selects "Borrow Book" in the system 8. The assistant scans the book's barcode/QR code 9. The system verifies book availability 10. The system records the borrowing transaction

	11. The system updates the book's status as borrowed 12. The system confirms successful borrowing to the assistant
Alternative Flow(s) - AF	<i>AF</i> AF1: Book Already Borrowed 3. The assistant scans a book that is currently borrowed 4. The system displays a warning and prevents the borrowing
Exception Flow(s) - EF	<i>EF</i> EF1: Scanner Malfunction 3. The scanner fails to read the barcode/QR code 4. The system prompts the assistant to retry or manually enter the book ID

UC012 Record Borrowing Transaction

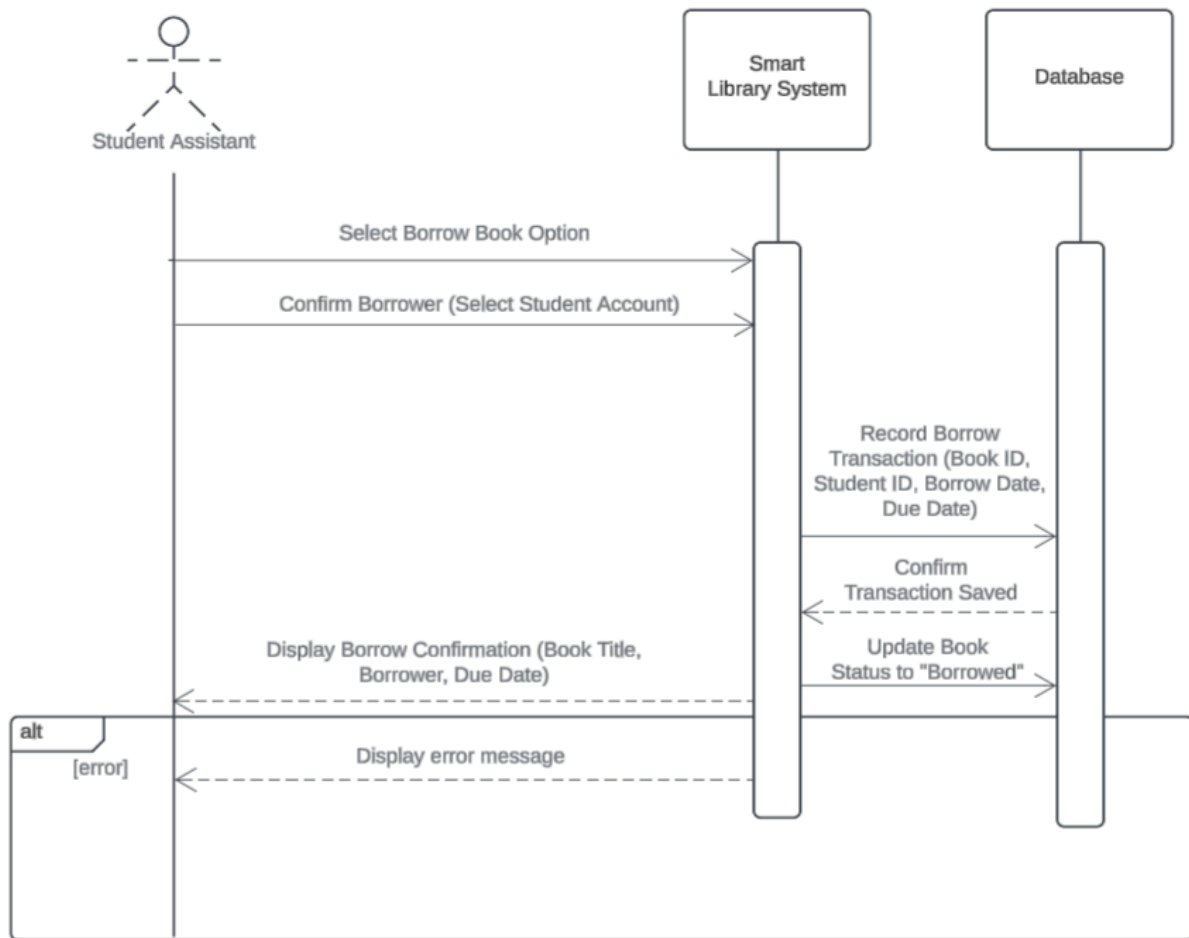


Figure 2.4.12.1: Sequence Diagram for Record Borrowing Transaction

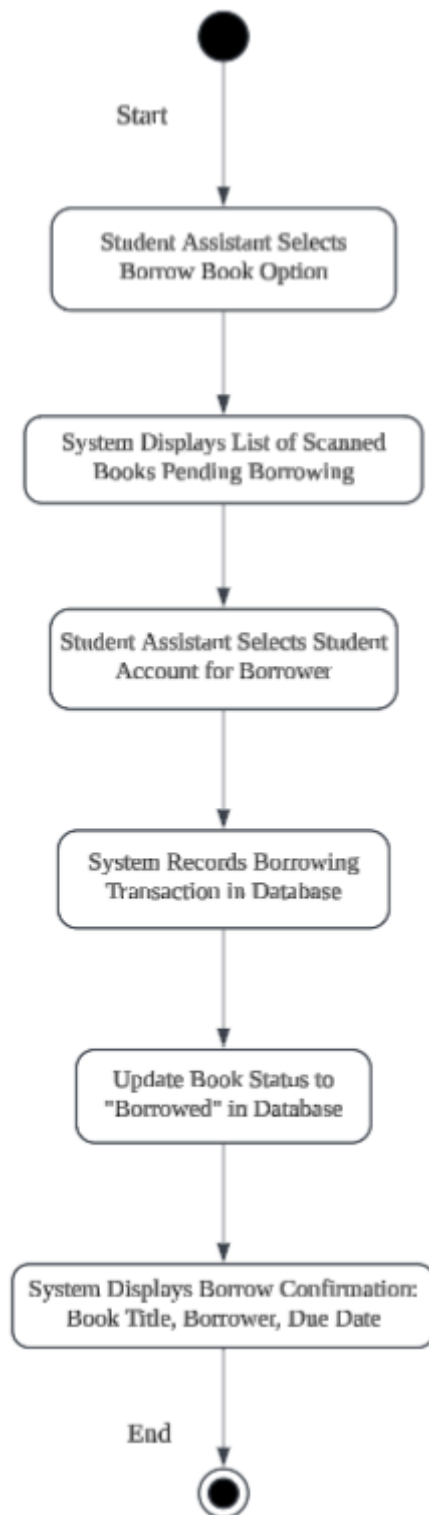


Figure 2.4.12.2: Activity Diagram for Record Borrowing Transaction

2.4.13 US013 User Story Update Book Availability Status

Table 2.13: User Story Description for Update Book Availability Status

User Story ID	US013
User Story Name	Update Book Availability Status
User Story Description	As a librarian or student assistant, I want the system to update the book status after a return is scanned, so that the book is marked as available and can be borrowed again.
Acceptance Criteria(s)	<i>Preconditions</i> • The user is logged in as Librarian or Student Assistant. • The book has an active borrowing record. • UC011 (Scan Book for Returning) has been successfully executed. <i>Postconditions</i> • Book availability status is updated in the database. • Borrowing transaction is closed. • Book is available for future borrowing or reservation. <i>Other Conditions</i> • If the book cannot be recorded due to a database error, the system should alert the student assistant • Borrowing history should be retained for audit purposes
Normal Flow(s)- NF	<i>NF</i> 1. The system receives the scanned book ID from UC011. 2. The system retrieves the active borrowing record for the book. 3. The system updates the borrowing record with the return date. 4. The system updates the book status to <i>Available</i>.

	5. The system confirms that the return process is successful.
Alternative Flow(s) - AF	<i>AF</i> AF1: Book Has No Active Borrowing Record 1. The system detects that the book is not currently borrowed. 2. The system displays a message indicating no active borrowing record.
Exception Flow(s) - EF	<i>EF</i> EF1: Database Update Failure 1. The system fails to update the borrowing or book status. 2. The system displays an error message and logs the failure.

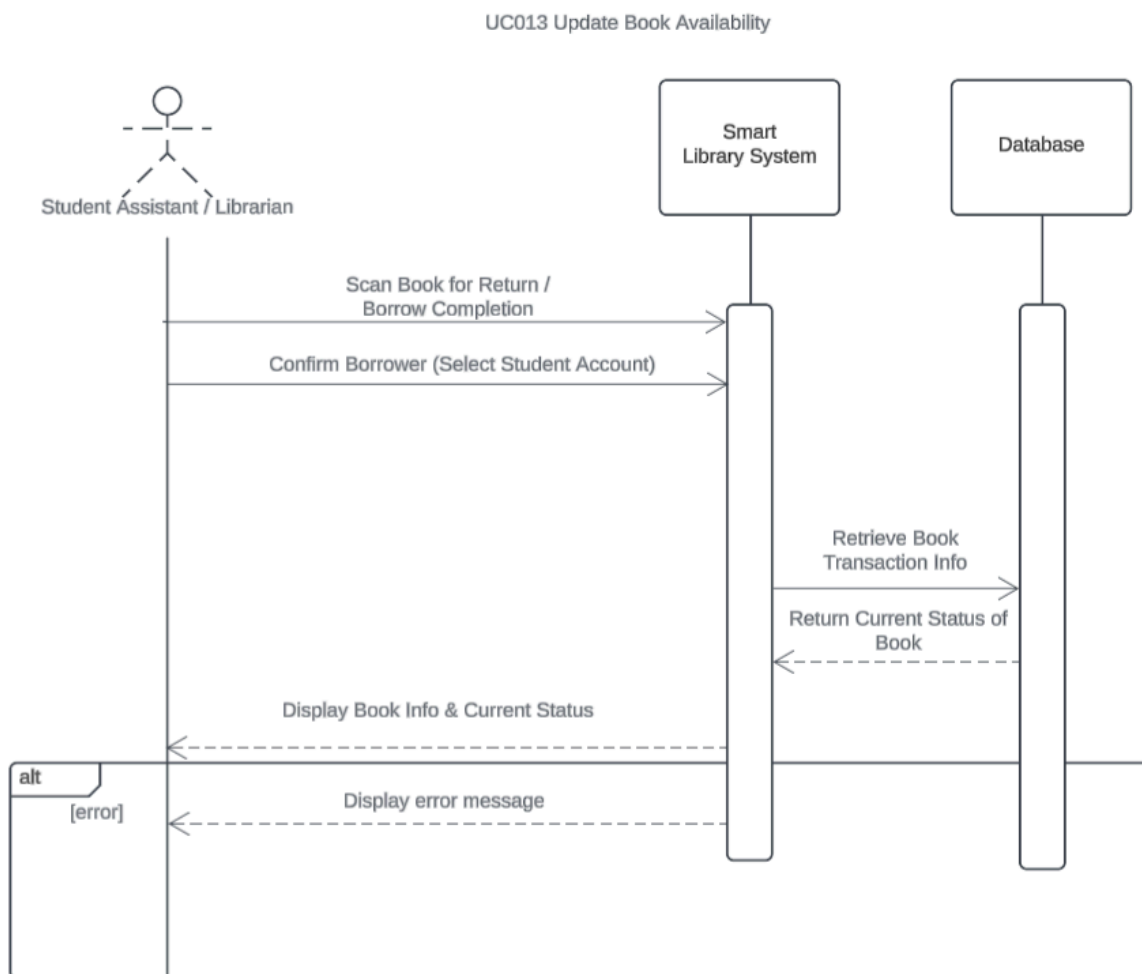


Figure 2.4.13.1: Sequence Diagram for Update Book Availability

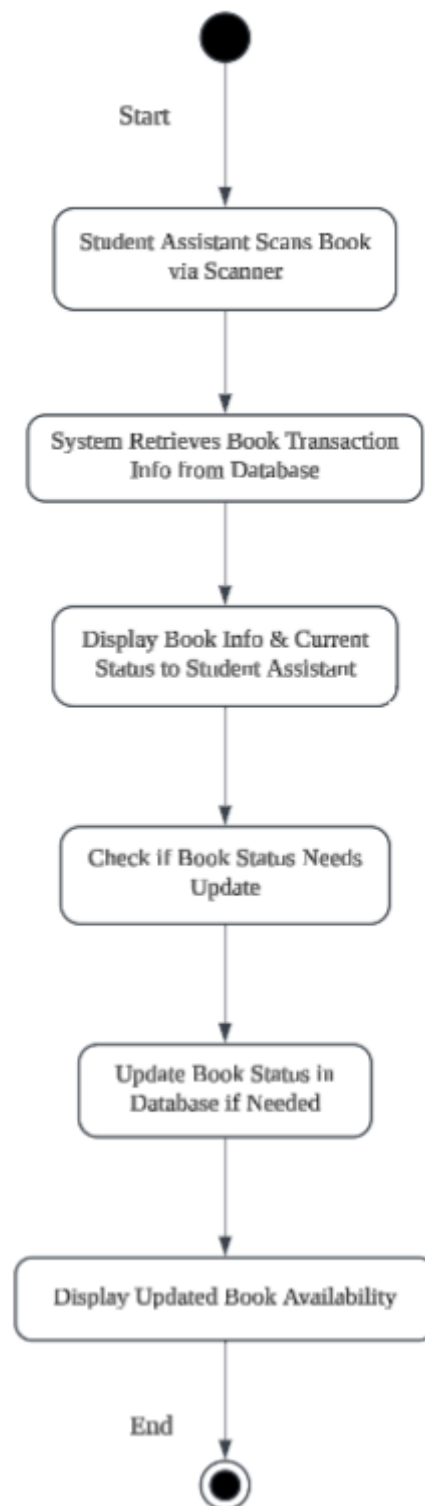


Figure 2.4.13.2: Activity Diagram for Update Book Availability

2.5 Performance and Other Requirements

This section specifies the non-functional requirements (NFRs) of the system. These requirements describe how the system should perform and operate, rather than specific functionalities. The requirements are categorized according to ISO/IEC/IEEE 29148, focusing on Performance Requirements and Software System Attributes.

Performance Requirements

Performance requirements define the system's ability to respond to user actions efficiently and reliably under expected operating conditions.

Response Time

- The system shall respond to user login requests within 3 seconds.
- Book search and filter operations shall return results within 2 seconds.
- OCR processing results shall be displayed within 10 seconds after image upload.
- Borrowing and returning transactions shall be completed within 3 seconds.

Throughput

- The system shall support at least 50 concurrent users performing standard operations (searching, viewing, borrowing).
- The system shall process a minimum of 30 OCR uploads per hour without performance degradation.

Capacity

- The system shall store at least **10,000 book records**.
- The system shall support at least **5,000 registered student accounts**.
- The system shall store digitized ledger records without data loss.

Availability

- The system shall be available 99% of the time, excluding scheduled maintenance.
- System downtime for maintenance shall be announced in advance.

Software System Attributes

Software system attributes describe the overall quality characteristics of the system.

Usability

- The system shall provide a simple and intuitive user interface suitable for librarians, student assistants, and students.

- The system shall minimize the number of steps required to perform common tasks such as borrowing and returning books.
- Error messages shall be clear and informative.

Reliability

- The system shall correctly record borrowing and returning transactions without data inconsistency.
- The system shall prevent duplicate or invalid transactions.
- The system shall recover gracefully from unexpected failures.

Maintainability

- The system shall be modular to allow easy updates and enhancements.
- System components such as OCR, catalog management, and user management shall be independently maintainable.
- The system shall provide logs for debugging and maintenance purposes.

Portability

- The system shall be accessible via standard web browsers.
- The system shall operate on common operating systems without requiring specialized hardware.

Compatibility

- The system shall integrate with OCR technology for digitizing ledger images.
- The system shall support barcode or QR code scanners for book borrowing and returning.

Security

- The system shall require user authentication before accessing protected features.
- User roles (Student, Student Assistant, Librarian) shall enforce access control.
- Sensitive data shall be protected from unauthorized access.

Safety

- The system shall not perform operations that could cause data loss without user confirmation.
- The system shall ensure that incorrect OCR data is verified before being saved.

Legal and Regulatory

- The system shall comply with applicable data protection and privacy regulations.
- User data shall be stored and processed in accordance with institutional policies.

Environmental

- The system shall reduce paper usage by digitizing physical ledger records.

- The system shall support environmentally sustainable digital record-keeping practices.

2.6 Design Constraints

Design constraints are limitations or conditions imposed on the system due to organizational policies, operational environment, hardware availability, security requirements, and external standards. These constraints must be considered during system design and implementation to ensure the system is deployable and acceptable within the organization.

Environmental Constraints

- The system shall be designed to operate in a library environment with standard indoor conditions.
- The system shall function correctly under temperatures ranging from 0°C to 40°C.
- The system shall operate reliably under typical indoor lighting conditions where barcode or QR scanners are used.

Hardware Constraints

- The system shall run on computers with a minimum of:
 - 8GB RAM
 - Dual-core processor
 - Minimum 256GB storage
- The system shall support the use of barcode or QR code scanners for book borrowing and returning.
- OCR functionality shall operate using standard document scanners or mobile camera devices.

Security Constraints

- The system shall enforce role-based access control for Librarian, Student Assistant, and Student roles.
- User authentication shall be required before accessing the system.
- Sensitive user and transaction data shall be stored securely and protected from unauthorized access.
- Only authorized users shall be allowed to modify book records and user accounts.

Compatibility Constraints

- The system shall be compatible with modern web browsers.
- The system shall be compatible with commonly used operating systems such as Windows-based systems used by the institution.
- The system shall be compatible with external OCR tools and scanning devices used by the organization.

Performance Constraints

- The system shall support at least 50 concurrent users without performance degradation.
- System response time for core operations (login, search, borrow, return) shall not exceed 3 seconds.
- OCR processing shall complete within an acceptable time frame to support daily library operations.

Organizational and Policy Constraints

- The system shall comply with the institution's library management policies.
- The system shall support enforcement of borrowing rules, due dates, and penalties as defined by the organization.
- Access to student borrowing records shall be limited to authorized personnel only.

Data Constraints

- The system shall store all book, user, and transaction records in a centralized database.
- Digitized ledger records shall be verified before being permanently stored.
- Data integrity shall be maintained during all borrowing and returning operations.

